Deep Learning for 3D Fashion Design: Survey From a Sewing Pattern-Driven Perspective

Jinbo Luo, Hong Qu, Yujie Zhao, Jie Zhang, *Member, IEEE*, and Yadie Yang

***bstract*—Sewing patterns form the structural foundation of the fashion industry, translating 2D conceptual designs into 3D manufacturable garments. With the rise of deep learning, 3D fashion design has seen transformative advancements, automating traditionally labor-intensive processes and enabling intelligent, data-driven workflows. Recent studies have demonstrated promising progress in areas such as sewing pattern generation, reconstruction, and 3D garment modeling. However, to date, no systematic review has specifically examined the integration of deep learning with sewing pattern–driven 3D fashion design. This survey addresses that gap by providing the first comprehensive overview of the field. We propose a novel four-stage pipeline, including representation, generation, reconstruction, and editing, to categorize and analyze current research. Within this pipeline, we review core methodologies, including geometric encoding for pattern representation, data-driven pattern generation, reconstruction from multimodal inputs (e.g., images, sketches, or text), and intuitive 3D garment editing techniques. We also consolidate existing benchmarks, covering both datasets and evaluation metrics, and contextualize the pattern-driven paradigm through comparison with alternative approaches in 2D and pattern-free 3D design. Finally, we identify key challenges, such as limited data availability and the di culty of incorporating domain-specific design constraints, and outline future research directions to address these issues. By synthesizing current developments and structuring the research landscape, this survey serves as a foundational resource to support and accelerate innovation in manufacturable sewing pattern-driven 3D fashion design.**

***Index Terms*—Sewing patterns, deep learning, 3D garment, 3D fashion design.**

I. Introduction

SEWING patterns are technical blueprints for garment construction, consisting of 2D geometric shapes that are cut from fabric and sewn together to assemble a 3D garment [1], [2], [3], [4]. This structural foundation makes them vital to both industrial and academic area, as the precise geometry of these panels directly determines the final shape, fit, and manufacturability of a garment [5], [6], [7], [8]. s illustrated in Figure 1, sewing patterns consist of essential components that form the foundation for transforming 2D designs into 3D garments. However, achieving the desired 3D outcome often requires designers to iteratively refine the pattern's shape and size, making this process both labor-intensive and time-consuming [9], [10], [11].

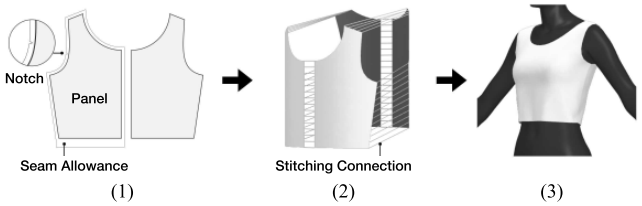


Fig. 1. Overview of the process from 2D sewing patterns to 3D garments. (a) drafting patterns with seam allowances and notches; (b) stitching corresponding panel edges; (c) assembling stitched panels into the final 3D garment.

The advent of digital 3D fashion design has further enhanced the significance of sewing patterns. Unlike purely 2D fashion design or mesh-based 3D fashion design approaches that primarily emphasize visual aesthetics, a sewing pattern-driven methodology, as implemented in industry-standard software like CLO3D and Style3D, ensures that digital garments align with the requirements of physical manufacturing [12], [13]. This approach ensures not only their physical realizability and manufacturability but also forms the link between virtual design and physical production. Recent advancements in deep learning have unlocked transformative potential for this pattern-driven workflow [14], [15], [16]. With its powerful capabilities in pattern recognition, geometric information processing, and generative modeling, these technologies offer new opportunities to automate, intelligently assist, and optimize the complex lifecycle of sewing patterns within the 3D fashion design pipeline [17], [18], [19], [20], [21], [22]. By integrating deep learning into this process, the fashion industry is poised to drive innovations in personalized customization and virtual fashion experiences, while simultaneously establishing a critical technical foundation for transitioning toward a more efficient and sustainable digital future.

Despite the great potential of deep learning, its systematic application within a sewing pattern-driven pipeline faces several critical gaps in the existing literature, underscoring the

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urgent need for a comprehensive review. First, while existing reviews have explored the intersection of deep learning and 3D fashion design, they generally provide a broader overview [15], [37], [38]. These studies review a wide range of generative I applications in fashion, covering images, 3D, and dynamic synthesis, but sewing patterns are often treated as one of many subtopics rather than as a fundamental element that connects digital design to physical production. Therefore, a systematic review from a specific sewing pattern-driven perspective is notably missing, which prevents us from understanding how deep learning can comprehensively address this critical issue. Second, the existing research on this topic is fragmented. These studies typically focus on isolated tasks such as generation or reconstruction, without providing a holistic view of the entire pipeline [39]. s a result, the field lacks a unified and cohesive pipeline that connects these disparate stages. Third, this field still lacks a comprehensive collection of high-quality datasets and standardized evaluation metrics [8], [30], [35], [36]. The fragmentation of these important resources makes it difficult for researchers to benchmark new methods and for the academic area to objectively evaluate progress.

To address these gaps, this review presents the first comprehensive overview of deep learning in sewing pattern-driven 3D fashion design. Our core goal is to help researchers gain a clear and systematic understanding of this field by proposing a novel multi-stage design pipeline (representation, generation, reconstruction, and editing). Within this pipeline, we integrate various research results and analyze the key methods and benchmarks in each stage. By structuring the existing knowledge, we aim to clarify the technical pipeline and highlight the interdependencies between research, thereby accelerating the development of more efficient digital fashion workflows.

The main contributions of this review are:

1) We comprehensively review the field from a sewing pattern-driven perspective, establishing a foundational reference that systematically connects deep learning techniques to the core principles of garment manufacturability. This provides the community with a clear roadmap to grasp the landscape of this critical sub-domain.
2) We structure the existing literature into a novel four-stage pipeline: representation, generation, reconstruction, and editing. For each stage, we provide an in-depth analysis of key methodologies and their interdependencies, transforming previously isolated studies into an integrated and understandable workflow.
3) We conduct a thorough consolidation of the field's benchmarks. This includes a systematic summary of available datasets and a detailed categorization of evaluation metrics, creating a centralized resource to facilitate standardized, reproducible research and accelerate future progress.

This review is based on a systematic search of literature from major academic databases, including IEEE Xplore, CM Digital Library, SpringerLink, ScienceDirect (Elsevier), and arXiv, focusing on publications from 2018 to 2025. We use a combination of keywords such as "sewing pattern", "3D garment", and "deep learning". This initial search yielded 252 candidate articles. Our selection criteria strictly prioritized research that directly links deep learning with manufacturable sewing patterns for 3D garment design, while excluding studies on pure 2D image synthesis or pattern-free 3D modeling. Following this screening process, 204 highly relevant articles were selected for in-depth analysis and inclusion in this survey, forming the foundation of our work.

II. Overview

This survey systematically explores the application of deep learning in sewing pattern-driven 3D fashion design. To provide a structured understanding of this domain, we organize our review around a comprehensive four-stage pipeline that serves as the backbone for this paper. s illustrated in Figure 2, this pipeline maps the entire digital garment lifecycle, with its core stages being:

(1) Representation focuses on transforming the intricate geometric and stitching details of sewing patterns into structured formats that can be fed into deep learning models. This section covers various representing schemes, from fundamental geometric descriptors to advanced tokenization and rasterization techniques, and examines their impact on model performance.

(2) Generation focuses on the computational creation of 2D sewing patterns to automate the initial design phase. This section covers both traditional rule-based methods and modern learning-based approaches, detailing the specific calculation formulas for the shape parameters of sewing patterns, as well as deep learning architectures like MLPs and Transformers.

(3) Reconstruction focuses on recovering 2D sewing patterns from various high-level inputs, including textual descriptions, 2D images, or sketches. This section reviews methodologies tailored to each input modality and discusses the challenges associated with multimodal information fusion.

(4) Editing focuses on modifying sewing patterns to optimize 3D garments and interactive techniques to support iterative design. This section covers two main approaches: precise editing for fine-grained control and fuzzy editing that uses intuitive inputs like natural language. We also highlight how deep learning facilitates these more intelligent and user-friendly design interactions.

To provide a systematic overview of how existing research fits within our proposed framework, we synthesize the state-of-the-art approaches in Table I. The table categorizes representative studies into distinct pipelines, where each row represents a unique workflow defined by its chosen methods for representation, generation, reconstruction, and editing. This classification not only organizes the current landscape but also serves as a roadmap for the detailed discussions in the subsequent sections.

The remainder of this paper is organized as follows. Section III discusses sewing pattern representation techniques. Section IV details sewing pattern generation methods. Section V explores sewing pattern reconstruction approaches. Section VI covers pattern editing methods. Section VII summarizes core datasets and evaluation metrics in this field. Section VIII presents common applications of deep learning in

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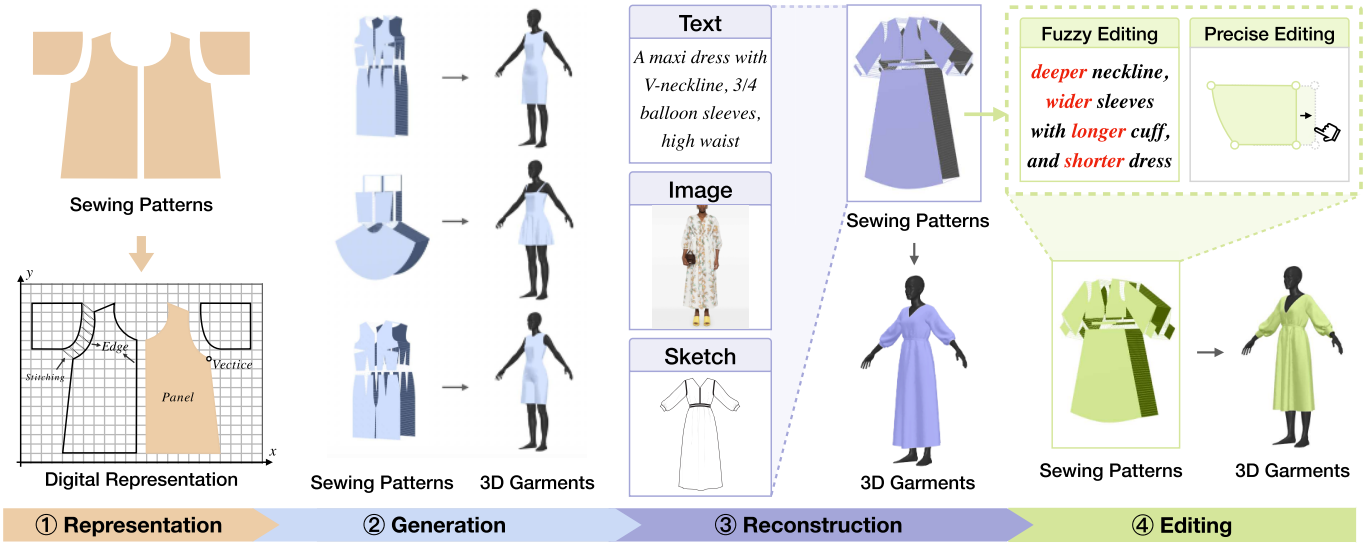


Fig. 2. General four-stage pipeline for sewing pattern-driven 3D fashion design: (1) **Representation** captures panel geometry (vertices, edges) and stitching information; (2) **Generation** maps high-level style features to structured various 2D patterns via generative models (rule-based or learning-based); (3) **Reconstruction** recovers patterns from multimodal inputs (text, images, sketches); (4) **Editing** interactively refines patterns via fuzzy or precise operations.

T BLE I
SUMM RIZ TION OF PIPELINES BOUT SEWING P TTERN-DRIVEN 3D F SHION DESIGN

Pipeline No.	Representation	Generation	Reconstruction	Editing	Datasets	Metrics	Studies
1	Dictionary	Rule-based	Text/Image/Sketch-based	Precise	Korosteleva & Lee; GarmentCodeData	-	[2], [8], [23]
2	Dictionary	Learning-based	Image-based	Fuzzy	Korosteleva & Lee	Average Error	[24]
3	Dictionary	Learning-based	Sketch-based	Fuzzy	Wang et al.; GarmentCodeData	#Panels, #Edges; Panel L2; Rot L2; Trans L2; Clothes-to-body Distance	[25]–[27]
4	Dictionary	Learning-based	Text/Image/Sketch-based	Fuzzy	GarmentCodeData	Simulation Success Rate; Aesthetic Score	[28], [29]
5	Tokenized	Learning-based	Text-based	Fuzzy	DressCodeData; GarmentCodeData-MultiModal	#Panels, #Edges; Panel L2; Rot L2; Trans L2	[30], [31]
6	Tokenized	Learning-based	Image-based	Fuzzy	SewFactory	#Panels, #Edges; Panel L2; Rot L2; Trans L2	[32], [33]
7	Tokenized	Learning-based	Text/Image/Sketch-based	Fuzzy	GarmentCodeData	#Panels, #Edges; Panel L2; Rot L2; Trans L2	[34]
8	Raster	Learning-based	Text/Image-based	Fuzzy	GarmentCodeData	IoU; CLIP Score	[35], [36]

sewing pattern-driven 3D fashion design. Finally, Section IX discusses limitations and future research directions, while Section X concludes the survey.

III. REPRESENT TION

Representation is the process of converting the raw information embedded in a sewing pattern, specifically its geometric outlines and stitching information, into a structured data format suitable for algorithmic manipulation [3], [33], [40], [41]. This representation underpins all digital workflows in 3D fashion design because it must capture both the 2D geometry of each panel and the relationships that govern how panels are joined. Geometric information is expressed through the coordinates of vertices and the boundary edges, including straight segments, Bézier curves, and B-splines, that enclose each panel [2], [8], [14]. Stitching information records sewing relationships that specify panel adjacency and alignment, such as the attachment of a sleeve cap to an armhole or notch-based seam-matching rules [5], [42], [43]. robust sewing-pattern representation therefore represents shape and structure in tandem, laying the foundation for downstream tasks such as generation, reconstruction, and interactive editing.

Geometric representation relies on parametric models, governed by mathematical formulas and adjustable parameters, to

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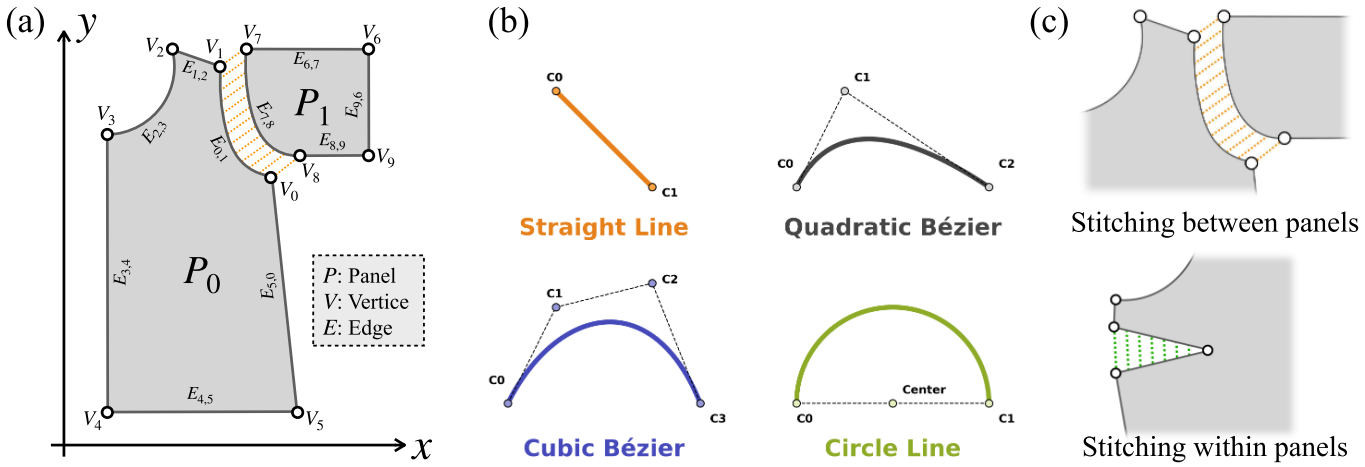


Fig. 3. The structural component representation for digital sewing patterns. (a) Geometric components; (b) Edge types; (c) Stitching types.

define sewing patterns and their stitching relationships. This structured representation underpins computational operations ranging from 2D pattern editing to 3D garment assembly [2], [8]. Such a representation defines the precise shape of curved edges using control points and mathematical equations, while it employs relational equations, such as linking shoulder slope to armhole depth, to enforce critical design constraints and ensure manufacturability [44], [45].

This section summarizes sewing pattern representation from two key perspectives. The first concerns the structural components that define a pattern's geometry and stitching information. The second addresses the representing methods used to translate these components into computationally amenable formats. Each of these aspects is detailed in the following sub-sections.

. *Structural Component Representation*

The structural component representation of a sewing pattern consists of four primary components: panels, vertices, edges, and stitches [2], [8], [32], [33], as shown in Figure 3(a). This structured approach is critical for digital fashion workflows as it enables the systematic representation of both geometric and stitching information.

(1) **Panels**: Panels are the fundamental components of a garment, representing 2D fabric shapes, such as the front and back pieces, sleeves, etc. Each panel is defined as a closed 2D contour: $Panels = \{P_0, P_1, \ldots, Pn\}(n = \text{number of panels})$. Each panel P_i is an independent closed polygon whose perimeter is formed by a cyclically ordered sequence of edges [14], [27], [46], [47], [48]. The shape of the panel is not a random polygon, but follows a priori rules of real pattern making. For example, panels may contain garment components such as darts, pleats or notches, which are associated with the panel border through the coordinate system or relative position.

(2) **Vertices**: Vertices are the atomic units of panel geometry, with their coordinates defining the panel's outline. The set of all vertices in a panel is represented as $Vertices = \{V_0, V_1, \ldots, V_k\}(k = \text{number of vertices})$, where each vertex V_m corresponds to a point (x_m, y_m) in a 2D coordinate system [2], [27], [31]. The precise location of these vertices ensures an accurate geometric representation of the panel. Moreover, adjusting these coordinates provides the primary mechanism for intuitive shape editing.

(3) **Edges**: Edges are the segments or curves that form the boundary of a panel by connecting pairs of vertices. The complete boundary is an ordered set of edges, denoted $Edges = \{E_{0,1}, E_{1,2}, \ldots, E_{m,n}\}$, where m, n are vertices' indices in the panel [2], [23], [27]. The representation of an edge is determined by its geometric type. Common types include straight lines as well as curves such as quadratic or cubic Bézier curves and circular arcs, as illustrated in Figure 3(b). For instance, a quadratic Bézier curve is expressed by the parametric equation $B(t) = (1 \quad t)^2 c_0 \quad 2t(1 \quad t)c_1 \quad t^2 c_2$, where c_0, c_1, and c_2 are the coordinates of its three control points and t is the parameter.

(4) **Stitches**: Stitches define the 3D assembly rules for a garment by specifying the connections between panels. These connections are represented as a set of paired edges, e.g., $Stitches = \{(E_{4,5}, E_{6,14}), (E_{7,8}, E_{8,0}), \ldots, (E_{m,n}, E_{m',n'})\}$.

critical constraint for a valid stitch is that the lengths of the paired edges must be equal, i.e., $Len(E_{m,n}) = Len(E_{m',n'})$ [2], [27], [35]. Stitches primarily connect edges between different panels, but they can also define internal seams within a single panel, such as a dart (Figure 3(c)(bottom)).

It is crucial to note that this idealized four-component model simplifies the complexity of real-world garments. It often overlooks essential non-panel components like pockets, plackets, collars, and cuffs, manufacturing details like seam allowances, and hardware placements like zippers, and buttons.

ddressing this gap between simplified digital representations and manufacturable patterns is a key challenge for the field [1], [3].

B. *Data Representation*

While a simple set of geometric primitives can capture the basic shape of a sewing pattern, this format is suboptimal for computational processing. s shown in Figure 4, to overcome this limitation, researchers have developed advanced

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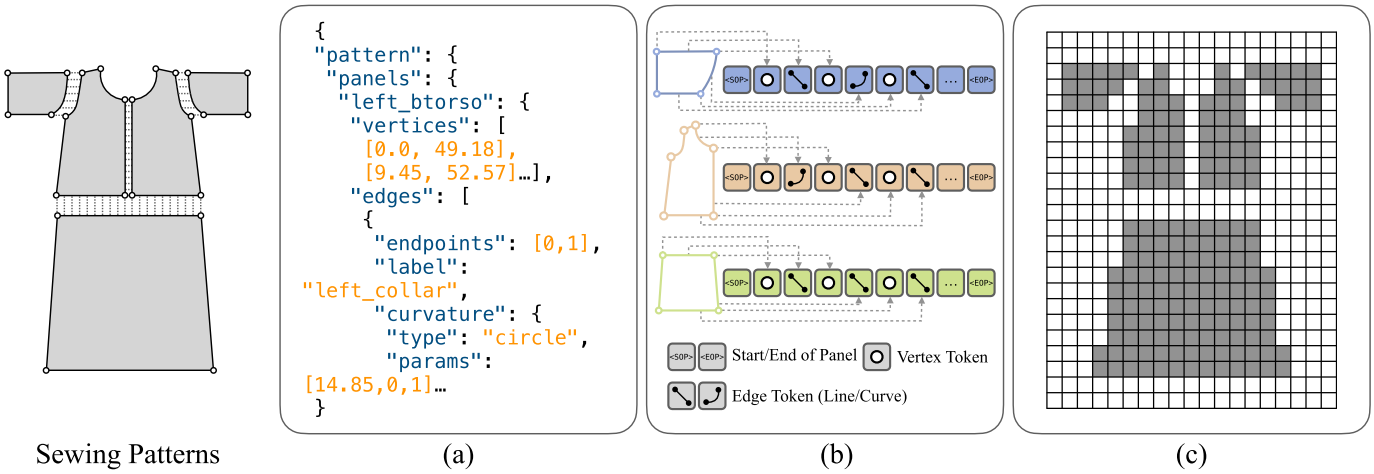


Fig. 4. Three data representation approaches for sewing patterns (left). (a) Dictionary representation [2], [8], [23], [29]; (b) Tokenized representation [30], [31], [34]; (c) Raster representation [35], [36].

representation methods that facilitate more efficient machine interpretation and manipulation of pattern data.

(1) Dictionary Representation: The dictionary representation is a prevalent method for capturing sewing pattern data. This approach utilizes a hierarchical key-value data structure to precisely capture both the geometry and metadata of sewing patterns, as illustrated in Figure 4(a). It is commonly implemented using JSON (JavaScript Object Notation), a format known for its excellent readability, scalability, and cross-platform compatibility [2], [8], [23], [29]. In this format, geometric parameters are typically defined using abstract units or relative ratios to represent 3D spatial relationships between elements. These abstract values are decoded into physical units (e.g., cm or mm) only during the reconstruction phase, where they are computed based on the target body measurements [8].

(2) Tokenized Representation: Tokenized representation involves quantizing sewing patterns into sequences of discrete tokens, a format highly amenable to modern I models. s shown in Figure 4(b), this approach facilitates effective model training and decoding by transforming complex geometric data into a sequential format. Methods such as DressCode, Ipparel, and GarmentDiffusion employ various tokenization schemes to enable I systems to interpret and process sewing pattern data more effectively [30], [31], [34]. To facilitate model processing, continuous geometric values are first normalized (e.g., to a [0, 1] range) and then quantized into integers using a scaling constant (e.g., multiplying by 1000 to create a vocabulary of discrete bins), as implemented in methods like DressCode [30].

(3) Raster Representation: The raster representation models sewing patterns as 2D grid-based bitmap data. Recent works, such as GarmentImage and GarmageNet [35], [36], have proposed this approach, which transforms sewing patterns into rasterized images comprising multiple concepts, including layers, internal/external markers, edge types, and local deformation matrices [49]. Standardized grid settings, such as 16 16 pixel resolution, are typically used to encode the topology and geometry of patterns in a scale-invariant latent space, which is subsequently mapped to physical dimensions during the decoding process. s shown in Figure 4(c), this method preserves the planar geometric information of the sewing patterns while maintaining the continuity of the latent space.

In summary, the advancement of sewing pattern representations has yielded significant benefits. It has led to more accurate 3D simulations by enabling high-fidelity geometric models that better reflect physical reality [3], [4], [50]. Furthermore, these robust frameworks are instrumental in empowering a new generation of I-driven tools [28], [30], [31], [41]. Having established how sewing patterns can be effectively represented, we now turn our attention to how they can be automatically generated.

IV. Gener tion

Generation of sewing pattern is the process of using computational technologies to translate high-level design intent and wearer-specific body dimensions into 2D patterns suitable for 3D simulation or physical fabrication [1], [2], [41]. Current approaches are broadly divided into two categories. Rule-based generation [2], [8], [23], [51], [52], [53], employs explicit mathematical formulations to define the geometry of 2D panels; learning-based generation [14], [24], [25], [27], [28], [29], [30], [32], [33], [35], [36], [40], [54], [55], leverages data-driven methods to generate sewing patterns in an end-to-end manner. Table III provides a summary and comparison of these key generation technologies.

. *Rule-Based Generation*

Rule-based methods generate sewing patterns by applying explicit geometric rules, such as the parameterization of 2D elements under mathematically defined constraints. Because designers can adjust geometric attributes directly, this paradigm provides strong controllability and clear interpretability. It is also independent of training data, making it well suited to situations where no prior examples are available. Within this framework, parametric techniques represent

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T BLE II
COMP RISON OF SEWING P TTERN TOKENIZ TION METHODS

Methods	DressCode [30]	AIpparel [31]	GarmentDiffusion [34]
Tokenization Unit	Single parameter value	Structured command token	Whole edge with all its parameters
Encoding Strategy	Full quantization of all continuous values, flattened into a long 1D sequence.	Discrete structure tokenization, with continuous geometric data handled separately.	Parameter embedding into a fixed-size grid of edge tokens.
Sequence Length	Very long ($>$ 30k tokens)	Very short (250 tokens)	Medium length (1k tokens)
Generation Model	GPT-like (Autoregressive)	Fine-tuned LMM (Autoregressive)	Diffusion Transformer (Parallel)
Generation Speed	Slow (minutes per pattern).	Fast (seconds per pattern)	Extremely Fast (sub-second per pattern)

T BLE III
SUMM RY ND COMP RISON OF SEWING P TTERN GENER TION TECHNOLOGIES

Methodology	Core Principle	Strengths	Limitations	Studies
Rule-based	Uses explicit mathematical formulas and parametric constraints to define pattern geometry.	• High degree of control and precision. • Interpretable and does not require training data. • Easily adaptable to specific body measurements.	• Labor-intensive to define rules for complex designs. • Limited creative scope beyond predefined rules. • Not scalable for generating highly diverse styles.	[2], [8], [53]
Learning-based (MLP)	Learns a direct mapping from a latent vector to a fixed-size vector representing the pattern.	• Simple architecture and computationally efficient. • Effective for garments with a fixed topology.	• Cannot handle variable numbers of panels or vertices. • Struggles to capture sequential or structural relationships.	[24], [25], [56]
Learning-based (LSTM)	Generates pattern boundaries as a sequence of vertices autoregressively.	• Naturally handles variable-length sequences (panel boundaries). • Captures local dependencies between adjacent vertices.	• Sequential nature leads to slow inference. • Struggles with long-range dependencies (e.g., symmetry).	[14], [33]
Learning-based (Transformer)	Processes entire sequences of pattern tokens in parallel using self-attention mechanisms.	• Excellent at modeling global and long-range dependencies. • Highly parallelizable, leading to faster training.	• Computationally expensive. • Requires large datasets to train effectively.	[30]–[32]
Learning-based (Diffusion)	Generates patterns by iteratively reversing a noising process, starting from random noise.	• Produces high-fidelity and highly diverse outputs. • Offers strong control through conditioning (e.g., text, images).	• Inference is computationally intensive and very slow. • Requires large datasets to train effectively.	[27], [35], [36]

a sewing pattern as a vector of geometric parameters, and modifying these parameters immediately alters the pattern's style and dimensions [2], [8], [53], [57], [58], [59], [60]. In 3D fashion design, this parameterization is implemented either through reusable template patterns or through parameter sets assigned to individual garment panels.

Several studies recover sewing-pattern parameters for a target 3D garment by iteratively optimizing a parametric template through physics-based simulation and geometric comparison [53], [59], [60]. In a typical workflow, a 3D scan of a person wearing the target garment is used as the reference geometry.

parametric sewing-pattern template is first draped over this body model in a physics-based simulator. differentiable solver or a simulated-annealing algorithm then iteratively refines the template parameters so that the simulated garment progressively matches the shape of the observed garment on the body, thereby minimizing their geometric discrepancy [61], [62], [63]. This inverse-design strategy is effective because the optimization not only aligns the shapes but also preserves wearability and physical realism. The process is driven by an objective function that typically minimizes the Chamfer distance between the simulated garment (S_{sim}) and the target garment (S_{tar}), defined as:

$$\mathcal{L}_{CD}(S_{sim}, S_{tar}) = \frac{1}{|S_{sim}|}\sum_{p\in S_{sim}} \min_{q\in S_{tar}} \|p \quad q\|_2^2 \\ \frac{1}{|S_{tar}|}\sum_{q\in S_{tar}} \min_{p\in S_{sim}} \|q \quad p\|_2^2 \qquad (1)$$

Recent advances in pattern making have produced techniques that adjust sewing-pattern parameters without the intermediate step of building a 3D human body model [2], [8], [23], [53], [57]. notable recent development is GarmentCode, which accommodates a broad range of panel geometries and accepts high-level inputs such as body measurements and style attributes [8], [23]. Its primary contribution is a domain specific language (DSL) for garment modelling that draws on object-oriented programming concepts [64]. This DSL

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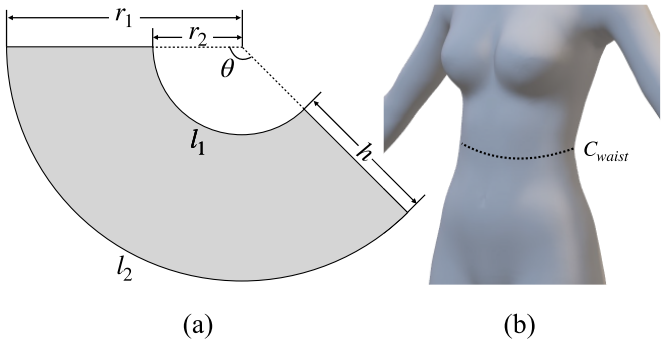


Fig. 5. Procedural skirt pattern parameterization. (a) Pattern geometry specified by top arc length l_1, bottom arc length l_2, top radius r_2, bottom radius r_1, skirt length h, and expansion angle θ. (b) Corresponding body input: waist circumference C_{waist}.

supports modular, fully parameterized pattern construction and allows designers to adjust garment characteristics with greater flexibility.

To illustrate the computational logic underlying such rule-based systems, we can examine the procedural generation of a skirt pattern, as depicted in Figure 5. The process starts with numerical inputs: a key body measurement, the wearer's waist circumference (C_{waist}), and two style parameters, a skirt length factor (P_{Length}) and a hem expansion parameters factor (P_{Suns}). These inputs are systematically transformed into the final geometric attributes of the 2D pattern through a cascade of calculations: $h = P_{\text{Length}} \cdot C_{\text{waist}}$, $\theta = P_{\text{Suns}} \cdot \pi$, $l_1 = C_{\text{waist}}$, $r_2 = l_1/\theta$, $r_1 = r_2 \quad h$, and $l_2 = r_1 \cdot \theta$.

This sequence of equations demonstrates how abstract inputs are translated into a complete set of concrete geometric values for the 2D pattern, ensuring it is tailored to both the wearer's body and the designer's specific stylistic intent. Due to its robust framework and clear procedural rules, GarmentCode and its associated datasets have been widely adopted in subsequent research [23], [27], [28], [29], [34], [54], [63].

B. Learning-Based Generation

Learning-based methods use deep neural networks to generate sewing patterns automatically by extracting implicit design rules from large-scale datasets [23], [43]. Unlike rule-based techniques, which rely on hand-crafted geometry, these approaches encode complex shape relationships and stylistic cues directly from data. Their common goal is to learn a probabilistic mapping from high-level design intent to concrete pattern geometry [25], [30], [40], [47]. ccording to network architecture, current work can be grouped into four categories: multi-layer perceptrons (MLPs) based models, long short-term memory (LSTM) based models, Transformer based models, and diffusion based models.

1) MLP-Based Models: Parameter estimation is the primary core task in sewing pattern generation. The task of mapping high-level inputs (e.g., body measurements or style vectors) to a fixed-size sewing pattern parameter vector is essentially a regression problem. variety of machine learning techniques, such as Support Vector Regression (SVR) have been effectively employed to predict garment parameters from 3D body scan data and user inputs [65]. Within the deep learning paradigm, MLPs often serve as the main or an auxiliary module. Because MLPs can learn non-linear functions that map structured inputs to parameter vectors, they suit problems in which a garment is described by a fixed set of values, for example panel dimensions, seam placements, or curvature values [14], [24], [25], [66]. Consequently, MLPs are typically employed to process structured inputs like body measurements or latent design vectors, learning a direct mapping to the corresponding sewing pattern parameters which, in turn, are used to construct the final garment shape.

Early works employed MLPs to encode an input and subsequently decode it into a fixed-length vector of pattern parameters, thereby generating the final pattern shape [24], [25], [56]. This process is framed as a regression task, where the model is optimized by minimizing a loss function that quantifies the discrepancy between the predicted and ground-truth parameters. The most common loss function for this purpose is the Mean Squared Error (MSE) [14], defined as:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N} \|\mathbf{p}_{\text{pred}}^{(i)} \quad \mathbf{p}_{\text{true}}^{(i)}\|_2^2 \tag{2}$$

where, $\mathbf{p}_{\text{pred}}$ is the parameter vector predicted by the MLPs, and $\mathbf{p}_{\text{true}}$ is the ground-truth vector. The loss penalizes the squared Euclidean distance between these vectors, averaged over a batch of N samples. While computationally inexpensive and effective for garments with a fixed parameter space, this approach cannot handle variable topologies, such as garments with a different number of panels or vertices, and struggles to capture the sequential nature of panel boundaries [24], [25], [56]. This limitation provides an improvement way for recurrent architectures that are more suitable for sequence modeling.

2) LSTM-Based Models: ddressing the inability of MLPs to model variable-length sequences, LSTM networks are particularly suitable for sewing pattern generation because patterns naturally consist of sequential structures, such as ordered sets of vertices or edges [69]. Unlike MLPs, which process input features independently, LSTMs maintain an internal memory state that captures dependencies across sequence elements. This capability makes them effective for modeling garment panel boundaries, which inherently form ordered sequences of vertices or control points. By processing these sequences autoregressively, LSTMs leverage preceding steps to inform subsequent predictions, thus facilitating the creation of sewing patterns with intricate boundaries or detailed panel geometries.

pproaches such as Panelformer and NeuralTailor have successfully employed LSTMs to improve the pattern generation process [14], [33]. In these methods, an LSTM module decodes a latent representation of the garment into a sequence of vertices representing panel boundaries. During training, the model is optimized to produce vertex sequences closely matching the ground-truth data. The loss function typically aggregates the L_2 distance between predicted vertex coordinates and corresponding ground-truth coordinates at each time

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step [33], [36]:

$$\mathcal{L}_{\text{seq}} = \frac{1}{T}\sum_{t=1}^{T} \|\mathbf{v}_{\text{pred},t} \quad \mathbf{v}_{\text{true},t}\|_2^2 \tag{3}$$

where T is the sequence length, $\mathbf{v}_{\text{pred},t}$ represents the predicted vertex coordinates, and $\mathbf{v}_{\text{true},t}$ denotes the ground-truth vertex coordinates at time step t. This sequential approach enables more flexible panel designs than fixed-parameter MLP models. However, a limitation of LSTMs is their difficulty in capturing long-range dependencies due to their sequential nature, making it challenging to enforce global properties like panel symmetry [70]. Furthermore, their autoregressive process can be slow and prone to error propagation in complex patterns.

3) Transformer-Based Models: To overcome the sequential processing and long-range dependency limitations of LSTMs, Transformer-based models have emerged as a powerful alternative [70]. Their ability to process sequence elements in parallel via self-attention makes them highly effective at modeling global dependencies within sewing patterns. This architectural advantage makes them superior for generating complex, multi-panel garments where understanding the global layout is critical [30], [31], [32].

widely adopted approach in this domain is Transformer-based autoregressive generation, which is especially suitable for predicting sequences of discrete tokens. Methods such as DressCode and Ipparel utilize this strategy by initially converting sewing pattern components, including panels, edges, and control points, into token sequences [30], [31], [67], [68]. These tokenized sequences serve as inputs to the model. During training, the model learns to predict the next token based on preceding tokens by optimizing a cross-entropy loss [31], [35], [36], [71]:

$$\mathcal{L}_{\text{CE}} = \sum_{t=1}^{T} \log P(\tau_t = \tau_{\text{true},t} | \tau_{<t}; \theta) \tag{4}$$

where $\tau_{\text{true},t}$ represents the true token at position t, $\tau_{<t}$ denotes the tokens preceding it, and θ are the model parameters. During inference, the Transformer employs autoregressive sampling to generate token sequences, which are subsequently decoded into the final sewing pattern geometry. While providing high structural fidelity, this capability comes at a significant computational cost, as the self-attention mechanism has a complexity quadratic to the sequence length. Transformers need more data than their recurrent models to achieve strong performance [72].

4) Di usion-Based Models: Owing to their superior these generative quality and diversity, Diffusion-based models have emerged as advanced frameworks capable of producing high-fidelity outputs through an iterative denoising process [73], [74]. Recently, this paradigm has been effectively adapted to sewing pattern generation tasks, offering both strong control and high fidelity in the resulting patterns. t its core, a diffusion model trains a denoising network ϵ_θ to estimate the noise ϵ that was added to an original clean data sample $\mathbf{x}_0$ at a particular timestep t. Specifically, a noisy data sample x_t is obtained through a forward noising process: $\mathbf{x}_t = \sqrt{\bar{}_t}\mathbf{x}_0 \quad \sqrt{1 \quad \bar{}_t}\epsilon$.

The training objective aims to minimize the difference between the predicted and the actual noise, typically employing a simplified loss function expressed as follows [27], [73], [75]:

$$\mathcal{L}_{\text{DM}} = \quad _{\mathbf{x}_0,\epsilon\sim \quad (0,\mathbf{I}),t} \quad \|\epsilon \quad \epsilon_\theta(\mathbf{x}_t, t)\|_2^2\big] \tag{5}$$

where the expectation is computed over clean data samples x_0, Gaussian noise samples ϵ, and time steps t.

notable innovation within diffusion models is the Diffusion Transformer (DiT) [76], which utilizes a Transformer-based denoising network in place of the traditional U-Net architecture.

Methods such as SewingLDM employ DiT-based latent diffusion techniques to generate sewing patterns that vary in size and complexity [27], [77]. SewingLDM typically follows a two-stage training procedure: initially, a base model is trained solely using text prompts; subsequently, it is fine-tuned by incorporating knowledge of body shapes and garment sketches, thereby enhancing its ability to generate patterns tailored to diverse body types and detailed garment structures [27].

nother relevant approach is GarmageNet, which introduces a diffusion-based framework for translating natural language descriptions or unstructured sewing patterns into structured, high-fidelity Garmage representations [35], [73]. GarmageNet employs two distinct Transformer-based denoisers: $_T$ focuses on the topological denoising stage to reconstruct global structural information such as 3D bounding boxes and panel scaling, while $_L$ is dedicated to the geometric denoising stage, utilizing the previously inferred topological data to accurately reconstruct detailed 3D geometry for each panel. The key trade-off for this state-of-the-art output quality is a substantial increase in computational cost. The iterative sampling process makes inference orders of magnitude slower than single-pass architectures like Transformers, posing a significant challenge for real-time and interactive applications [78].

V. Reconstruction

Reconstruction refers to the process of recovering 2D sewing patterns from high-level garment representations, such as textual descriptions, images, or sketches. This task aims to infer the underlying manufacturable panels corresponding to an abstract visual or descriptive garment concept [30], [32], [79], [80]. This capability is particularly valuable to fashion designers, whose design processes often begin with conceptual representations and who require effective techniques to translate these abstract ideas into production-ready sewing patterns [79], [81], [82]. core challenge stems from the inherent ambiguity of these high-level inputs. single sketch or description can correspond to multiple valid garment designs, creating a one-to-many mapping problem [25], [28]. Rather than modeling this diversity, current deep learning approaches predominantly handle this issue by framing it as a deterministic, one-to-one prediction task [30], [32]. This is a direct consequence of their training objectives. By optimizing regression-based losses, such as L2 distance, models are mathematically encouraged to predict the average of all possible

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T BLE IV
SUMM RY ND COMP RISON OF RECONSTRUCTION TECHNOLOGIES B SED ON INPUT MOD LITY

Input Modality	Core Challenge	Key Techniques	Strengths of Modality	Studies
Text	Translating abstract, high-level semantic descriptions into precise, quantitative geometric parameters for sewing patterns.	• Text Encoders: CLIP, T5 • LLMs: LLaMA, LLaVA	• Highly expressive for abstract concepts and styles. • Intuitive and accessible for non-expert users. • Enables fine-grained control via detailed prompts.	[27]–[31]
Image	Disentangling garment geometry from complex visual information such as human pose, occlusion, lighting, and texture.	• CNNs: U-Net, ResNet. • Hybrid: CNN feature extractor + Transformer decoder • Multimodal LLMs: LLaVA (CLIP-ViT)	• Provides rich, detailed visual reference. • Directly digitizes existing physical garments.	[24], [32], [33], [36]
Sketch	Inferring precise geometric structure and constraints from abstract, often geometrically imprecise and sparse line drawings.	• Sketch Encoders: sketchRNN • Data Synthesis: Using models like PiDiNet • Multimodal LLMs: LLaVA (CLIP-ViT)	• Natural and direct medium for initial design idea. • Unconstrained by real-world visual complexities.	[25]–[28]

valid outputs for a given input. Consequently, they implicitly learn to generate a single pattern that represents the central tendency within the training data, rather than the full spectrum of creative possibilities.

The reconstruction task encompasses several specialized sub-problems categorized by their input modality: text-based, image-based, and sketch-based reconstruction. Each type of input modality introduces distinct technical challenges that necessitate dedicated methodological solutions. This section systematically reviews state-of-the-art approaches for each input category and discusses the principal innovations facilitating the automated reconstruction of sewing patterns. comparison of these reconstruction approaches based on input modality is presented in Table IV.

. *Text-Based Reconstruction*

Reconstructing precise sewing patterns from textual descriptions is a challenging task that sits at the intersection of natural language processing (NLP) and garment engineering [83], [84]. While text descriptions effectively articulate high-level design features, such as "a sleeveless -line dress with a scoop neck, knee-length," they typically lack the quantitative geometric data required for pattern creation, including precise measurements and construction details. Consequently, recent research has focused on developing systems capable of inferring these technical specifications from abstract semantic inputs.

Significant progress has been made in generating sewing patterns from natural language by employing advanced text-processing techniques. prominent example is DressCode [30], which introduced SewingGPT, an autoregressive Transformer model explicitly tailored for this purpose. SewingGPT first utilizes a pre-trained CLIP text encoder to convert garment descriptions into semantic feature tokens [85], [86]. These tokens are then projected into a conditional embedding vector h using an MLP before being input to a Transformer decoder. By combining CLIP embeddings with cross-attention mechanisms, SewingGPT successfully captures subtle semantic details in textual prompts and translates them into structured sewing patterns.

lternative methods have adopted different pre-trained language models for text encoding. For example, SewingLDM [27] leverages the T5 model, which employs an encoder-decoder architecture to encode text into semantic tokens, such as words or subword units [70], [87]. More recently, approaches favoring the computational efficiency of decoder-only architectures have emerged. Systems like ChatGarment, Design2Garment, and Ipparel exemplify this trend by building upon open-source models such as LLaM and LLaV [28], [29], [31], [88], [89]. These models typically follow a consistent text-processing pipeline: they initially tokenize input text into subword units using a SentencePiece tokenizer based on the Byte-Pair Encoding (BPE) algorithm [90], and subsequently map these tokens to vector embeddings as inputs to following Transformer layers.

B. *Image-Based Reconstruction*

Reconstructing precise sewing patterns from images is a complex undertaking at the intersection of computer vision (CV) and domain-specific garment engineering [33], [91], [92], [93]. Unlike text, images from sources like runway photos or product catalogs offer rich visual information, capturing nuanced attributes such as silhouettes, texture details, and 3D draping. However, the computational complexity involved in processing visual data is significantly greater than that of text-based data [94], [95], [96]. Therefore, recent research focuses on developing methods to create geometrically accurate and production-ready sewing patterns from images, thereby facilitating the rapid digitization of physical garments.

Significant advancements have been achieved in image-to-pattern reconstruction, with modern methods effectively tackling challenges such as complex human poses and occlusions. Foundational approaches have primarily relied on Convolutional Neural Networks (CNNs) and their variants. For instance, SPnet [24] utilizes a multi-stage CNN pipeline, first normalizing input images of dressed individuals

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into a standardized T-pose representation via a convolutional encoder-decoder. Subsequently, this standardized view is passed to a second stage consisting of a convolutional autoencoder combined with an MLP to predict the corresponding sewing pattern parameters [97]. Similarly, GarmentImage [36] employs a sequential pipeline based on multiple U-Net models, progressively inferring garment labels, edge types, and deformation matrices from the input image [98]. More recent work has shifted towards hybrid architectures that combine the strengths of CNNs and Transformers. Methods such as Sewformer and Panelformer adopt a two-step pipeline, initially extracting features through a ResNet50 backbone, followed by a Transformer module designed to capture global structural dependencies [32], [33], [70], [99], [100], [101], [102]. Extending this hybrid concept, Dress-1-to-3 [55] incorporates an additional Transformer encoder stage that refines the visual embeddings, thereby enhancing pattern-generation quality.

Beyond traditional CV methodologies, multimodal large language models (LLMs) are emerging as a powerful alternative for image-based reconstruction. This approach is exemplified by systems like ChatGarment and Ipparel, which leverage LLaV for joint text and image understanding [29], [31], [88]. LLaV typically employs a vision transformer, such as CLIP-ViT or SigLIP, as its image encoder. This encoder processes an image by dividing it into patches (e.g., 14 14) and transforming them into a sequence of visual tokens, which are then integrated with textual embeddings to enable comprehensive multimodal reasoning [85], [103]. Other systems adopt similar but distinct multimodal strategies. Design2Garment, for instance, utilizes the proprietary GPT-4V model for its visual encoding capabilities [28]. In another variation, GarmentX [54] employs the DINOv2 ViT-B/14 model to extract conditional tokens from the image. These tokens subsequently guide a Masked utoencoder (M E) and a diffusion-based MLP to generate the final output, demonstrating a sophisticated fusion of different generative modeling paradigms [104], [105].

C. Sketch-Based Reconstruction

Sketch-based reconstruction is the task of automatically generating sewing patterns from design sketches. This capability is highly valuable as sketches are a primary medium for fashion designers to articulate their initial concepts, capturing key features like silhouette, style lines, and intended fabric drape [82], [106], [107], [108], [109]. The core challenge in this area arises from the intrinsic characteristics of sketches: while they effectively convey semantic intent, sketches are often abstract and geometrically imprecise, lacking the precise dimensional information necessary for manufacturing [25], [110], [111]. lthough less extensively explored than image-based reconstruction, several notable approaches have emerged to bridge the gap between abstract design concepts and precise, manufacturable outputs.

Recent studies are exploring the direct generation of sewing patterns from sketches through diverse computational techniques. For example, PersonalTailor [26] employs sketches as high-level guidance for interpreting existing 3D garment point clouds. Specifically, the method predicts 2D panel shapes and sewing details by conditioning the generated output on a latent vector extracted from sketches using sketchRNN [112]. Notably, this latent vector encodes conceptual panel characteristics rather than literal sketch geometry.

Other approaches utilize multimodal learning techniques for more direct sketch-to-pattern translation. SewingLDM [27] processes sketches in a manner analogous to images. To overcome the common challenge of data scarcity, it innovatively creates its training data by leveraging the PiDiNet model to synthesize sketches from the GarmentCodeData image collection, thereby enriching the diversity of its multimodal training set [23], [113]. However, these sketches differ from those in real-world sketches in terms of structural complexity. Consequently, the model's performance tends to decline when generalizing to these real-world sketches.

VI. Editing

Editing refers to the modification of existing sewing patterns within computer-aided 3D fashion design systems. It encompasses a range of operations that alter garment structure, shape, or dimensions based on user input [114]. central challenge in this stage is not just enabling modifications, but ensuring that these changes maintain the design's overall integrity and coherence. For instance, a simple command like "make the sleeves longer" must be translated into a geometrically valid adjustment that is also aesthetically pleasing and proportional to the rest of the garment [50], [115]. This involves complex considerations of constraint satisfaction and style preservation, which are critical for producing manufacturable and well-designed outcomes. Depending on the interaction mode and required level of precision, editing techniques are typically categorized into two types. s shown in Figure 8, precise editing allows fine-grained control through parametric adjustments or direct manipulation of pattern components [50], [115]. On the other hand, fuzzy editing interprets high-level, intent-driven commands from natural language or sketches into specific modifications of sewing patterns [28], [29]. These two editing paradigms are summarized and compared in Table V.

. Precise Editing

Precise editing refers to the ability to modify specific garment components with high accuracy, typically through direct operations such as dragging vertices or adjusting parameters [8], [10], [57], [114]. This process relies on a clearly defined structural representation of sewing patterns, often expressed in geometric or parametric form, which enables accurate and localized modifications. Two primary methods are commonly used in precise editing: geometric control and parametric control.

Geometric control, as shown in Figure 8(a), allows users to directly manipulate geometric elements, such as control points or boundary lines, on either the 2D patterns or the 3D model. Changes made in one view are automatically synchronized with the other in real time. Sensitive Couture [114] is a representative example of this approach, pioneering a system that maintains a consistent correspondence between 2D pattern

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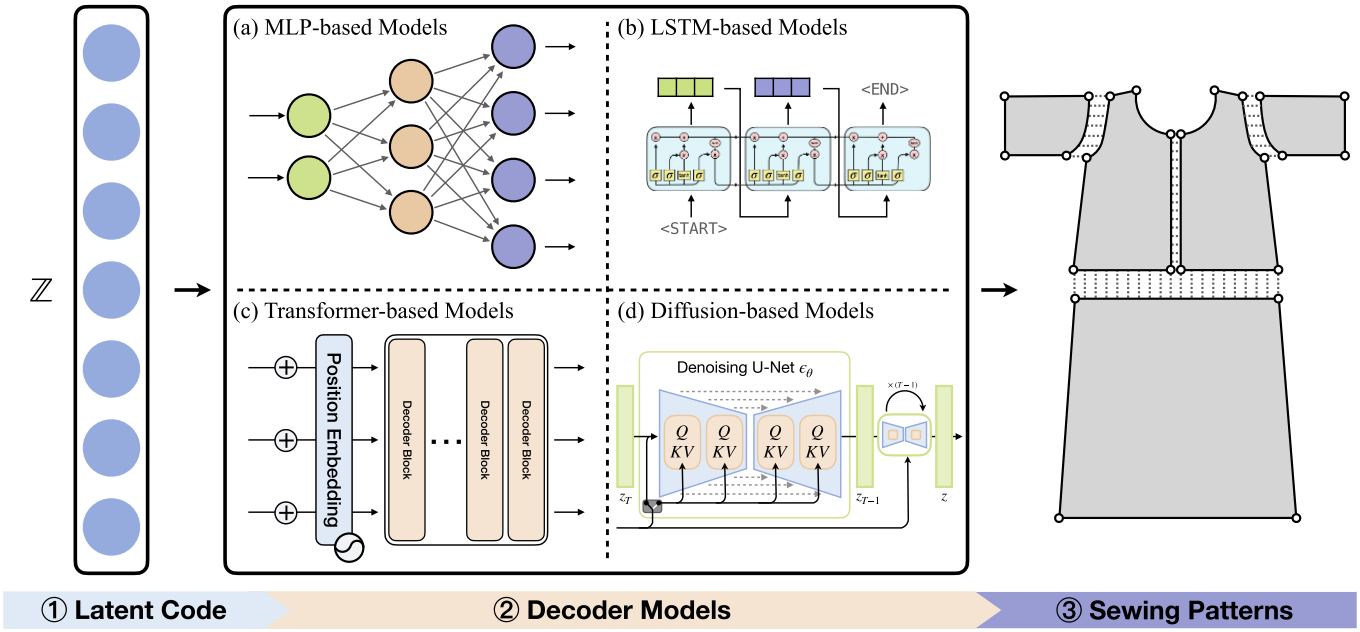


Fig. 6. Learning-based pipeline for generating sewing patterns. latent code $\mathbb{Z}$ (1), encoding garment properties, is input to a decoder model (2) that generates the 2D sewing pattern (3). Four primary decoder models are explored: (a) MLP-based models [24], [25], [56]; (b) LSTM-based models [14], [33]; (c) Transformer-based models [30], [31], [67], [68]; and (d) Diffusion-based models [27], [35].

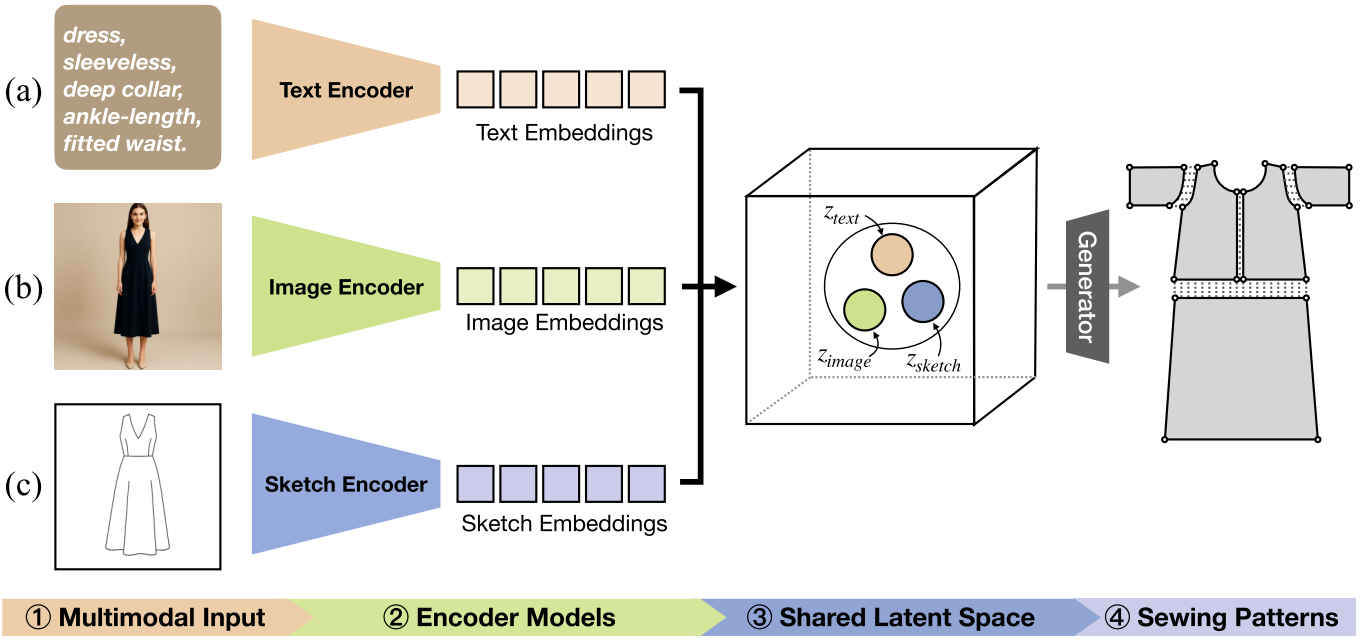


Fig. 7. Multimodal data parsing framework. (a) Text input [27], [28], [29], [31], (b) Image input [24], [27], [29], [31], [32], [33], [36], [55], and (c) Sketch input [26], [27], [28]. This framework consists of four main steps: multimodal input, encoder model, shared latent representation, and sewing patterns generation.

edits and the resulting 3D draped garment [10], [57]. This is achieved through a combination of sensitivity analysis for 3D shape prediction, dynamic re-meshing for boundary adaptation, and advanced physical simulation to ensure stable garment deformation [116]. Building on these capabilities, other systems provide more specialized tools. For instance, PerfectTailor supports a variety of 3D editing operations such as "cut", "extend", and "tighten" [115], while ISP enables precise edge modification by allowing users to draw or erase lines directly on the 2D panel [48].

Parametric control enables users to adjust preset parameters, such as sleeve length, waist circumference, or tightness weight, through parameter sliders or input boxes to control the overall or local shape of the garment. GarmentCode exemplifies this approach by modeling clothing design using an object-oriented paradigm [8], [23]. Users can modify key attributes such as

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T BLE V
SUMM RY ND COMP RISON OF SEWING P TTERN EDITING TECHNOLOGIES

Editing Type	Interaction Method	Strengths	Limitations	Studies
Precise	Direct manipulation of geometric elements (vertices, curves) or adjustment of numerical parameters (e.g., sliders for length, width).	• Offers fine-grained, exact control over the pattern geometry. • Deterministic and immediate result.	• Users need to understand pattern making principles.	[8], [48], [57], [114], [115]
Fuzzy	Using high-level, abstract inputs like natural language commands or rough sketches.	• Aligns with a designer's natural workflow and communication style. • Lowers the barrier for users without deep pattern-making knowledge.	• Natural language can be underspecified and lead to misinterpretation. • Difficult to achieve precise, specific geometric modifications.	[28], [29], [54]

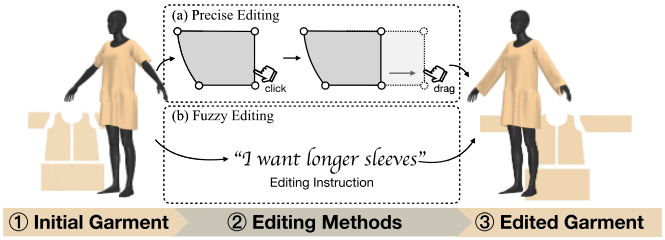


Fig. 8. Pipeline of editing for sewing patterns. (a) Precise editing [8], [57], [115], and (b) Fuzzy editing [28], [29]. This pipeline consists of three main steps: (1) Initial garment, (2) editing methods, and (3) edited garments.

trouser length or hip circumference, and the corresponding 2D patterns and 3D models update immediately to reflect the changes. Expanding on this framework, GarmentX [54] introduces the use of parameter vectors for localized adjustment. This enhancement allows users to alter specific garment regions independently, eliminating the need to regenerate the entire pattern and thereby improving the efficiency of iterative design.

B. Fuzzy Editing

Fuzzy editing refers to the use of high-level user interactions, most commonly natural language instructions, to modify garment designs. This method requires the system to interpret user intent and translate abstract verbal commands into specific alterations of the sewing patterns [28], [29]. s shown in Figure 8(b), by aligning with intuitive human communication, fuzzy editing lowers the barrier to conceptual design and facilitates rapid prototyping.

Meanwhile, this process presents significant challenges in maintaining design integrity. The core difficulty lies in translating ambiguous, qualitative commands into precise, quantitative edits while preserving garment style and proportionality. For example, the instruction "make the sleeves longer" is underspecified. naive geometric stretch would likely distort the cuff shape and disrupt the seam connection at the armhole. successful system must infer the designer's intent and apply a modification that respects the underlying design constraints [28], [29]. This includes adjusting the sleeve panel's length while potentially scaling the cuff, ensuring the armhole seam length remains compatible with the corresponding torso panel, and preserving the overall style of the sleeve.

Recent systems have begun to address this challenge by leveraging highly structured, parametric pattern representations. critical requirement for effective fuzzy editing is the use of structured and disentangled parameter representations, which allow for targeted modifications to specific garment components without affecting unrelated parts. This approach is demonstrated in systems such as Design2GarmentCode [28] and ChatGarment [29]. In Design2GarmentCode, a Multimodal Understanding gent (MMU) first interprets the user instruction, which is then translated into a parameterized GarmentCode [8] program through a DSL Generation gent (DSL-G). Similarly, ChatGarment employs the LLaV model to determine which garment parts are referenced in the command and utilizes the GarmentCode framework to adjust only the relevant parameters, thereby preserving the integrity of the remaining design. By modifying a high-level parameter like "sleeve_length" instead of manipulating raw vertices, these systems ensure that all related geometric features adjust coherently according to predefined design rules embedded within the GarmentCode structure [8]. These methods successfully preserve geometric validity and proportion, but there is still a lack of research on pleasing aesthetics, which represents an area worth exploring.

VII. BENCHM RKS

This section presents the datasets and evaluation metrics commonly used in sewing pattern-driven 3D fashion design. These benchmarks serve as standardized tools for training models, validating results, and comparing performance across studies. The datasets typically consist of annotated garment patterns, 3D draped models, and associated multimodal inputs, while the evaluation metrics assess the accuracy, realism, and usability of generated outputs. Together, they support consistent experimental protocols and facilitate objective comparisons among different approaches.

. Datasets

While benchmarks for pure 3D garment reconstruction, such as CLOTH3D [117] and MultiGarmentNet [118], provide valuable context for the broader field, this survey exclusively

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T BLE VI
SUMM RIZ TION OF KEY SEWING P TTERNS D T SETS

Dataset	Year	Scale	Modalities		
			Text	Image	Sketch
Wang et al. [25]	2018	8k	×	✓	✓
Korosteleva & Lee [2]	2021	23.5k	×	✓	×
SewFactory [32]	2023	19.1k	×	✓	×
DressCodeData [30]	2024	19.3k	✓	×	×
GarmentCodeData [23]	2024	115k	×	✓	×
GarmentCodeData-MultiModal [31]	2024	120k	✓	✓	×
GarmentX [54]	2025	378.7k	×	✓	×

focuses on datasets containing the explicit sewing pattern data essential for our manufacturability-driven analysis. Large-scale, high-quality datasets empower data-driven methods for sewing pattern generation and garment design by enabling models to learn the mapping between multimodal inputs and sewing patterns. Over the past few years, several benchmark datasets have been introduced, each tailored to support different tasks and modalities in the fashion pipeline [2], [23], [32]. These datasets provide paired multimodal data, including 2D sewing patterns, 3D garment meshes, text descriptions, images, and sketches, which are essential for training deep learning models across a range of applications. Table VI summarizes representative datasets widely used in this field.

n early dataset developed by Wang et al. [25] includes 8,000 samples for three garment categories: shirts, skirts, and kimonos. Each sample comprises a parameterized 2D sewing pattern, an image, and a sketch, encoded in a unified parameter space that captures geometric properties such as sleeve length and waist size, as well as material attributes. Building upon this, Korosteleva and Lee [2] constructed a larger dataset with 23,500 sewing patterns generated from 19 base templates. These templates cover a wider range of garment types, including pants and dresses, and allow scalable pattern generation through parameter variation.

Subsequent datasets have expanded reconstruction capabilities by focusing on different input modalities. To address the challenge of image-based reconstruction, SewFactory [32] introduced a dataset containing one million garment images paired with 19,100 sewing patterns. This dataset captures diverse poses and body shapes, enabling robust image-to-pattern learning. For text-based reconstruction, DressCodeData [30] includes 19,300 samples across 11 garment categories, each annotated with corresponding text descriptions and texture maps to support sewing pattern generation from text prompts.

GarmentCodeData (GCD) [23] provides a comprehensive dataset of 115,000 garment instances, each containing a complete set of annotations including 2D patterns, 3D meshes, UV maps, body measurements, and parametric design information. This dataset facilitates research on garment reconstruction, simulation, and personalization. Its extension, GarmentCodeData-MultiModal (GCD-MM) [31], adds paired textual and image data for 120,000 garments, as well as annotations for garment editing tasks. GarmentX [54], one of the most recent and extensive datasets, further scales this effort by offering 378,682 garment instances, each consisting of parameter-image pairs. It introduces a structured representation designed to support semantic editing. Critically, however, these large-scale datasets are procedurally generated from a limited set of parameterized templates [2], [23]. While their styles are scalable, this approach limits design diversity to variations within the underlying base patterns. Consequently, these datasets often lack complex, real-world styles, such as intricate necklines or cuffs [1], [3]. This leaves models trained on this data with a significant gap in generalization to more creative fashion domains.

B. Evaluation Metrics

Evaluating sewing pattern-driven systems requires a comprehensive approach, as it spans multiple stages of the design and generation pipeline. Key evaluation components include the accuracy of multimodal input reconstruction, the geometric precision of the generated 2D sewing patterns, and the visual and physical realism of the resulting 3D garments [14], [29], [30], [32]. Each of these components presents unique challenges and calls for specialized metrics. Table VII outlines the primary evaluation criteria commonly adopted in this research domain.

1) Metrics for Sewing Patterns: The evaluation of sewing patterns focuses on their fundamental geometric and stitching information. These metrics are designed to directly compare the predicted 2D panels with ground-truth data, assessing the accuracy of panel counts, shapes, and their critical sewing relationships [14], [32]. SPnet directly defines numerical parameters for sewing patterns and calculates the average error between ground-truth values and predicted parameters to estimate the quality of the model [24]. NeuralTailor introduced several foundational metrics for this purpose. To evaluate structural accuracy, it uses the number of panels (#Panels) and the number of edges (#Edges). To assess geometric precision, it computes the average L2 distance between predicted and ground-truth panel vertices (Panel L2), as well as the L2 distances for panel rotation (Rot L2) and translation (Trans L2). To facilitate direct L2 comparison, panel positions are defined in a reference frame anchored by consistent semantic points, which inherently aligns the predicted and ground-truth poses without the need for an explicit alignment step [14].

dditionally, Precision and Recall are employed to evaluate the correctness of predicted stitch connections. These metrics have been widely adopted in subsequent studies [31], [32], [34], [35].

Further refinements to pattern quality evaluation have been introduced by recent research. GarmageNet [35] utilizes the verage Matching Distance (MD), which measures the average Euclidean distance between corresponding vertices along stitched edges after optimal alignment. GarmentImage [36] applies the Intersection-over-Union (IoU) metric, calculated by rasterizing the predicted and ground-truth panels onto

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T BLE VII
SUMM RIZ TION OF KEY EV LU TION METRICS FOR SEWING P TTERN-DRIVEN 3D F SHION DESIGN

Category	Metric	Description	Studies
Sewing Patterns	**Average Error**	Average error of predicted parameter value from ground truth.	[24]
	#Panels, #Edges	Correctness of panel and edge counts.	[14], [31], [32], [34]
	Panel L2	Accuracy of vertex position between predicted and GT panels.	[14], [31], [32], [34]
	Rot L2, Trans L2	Accuracy of panel rotation and translation.	[14], [31], [32], [34]
	Stitch Precision/Recall	Accuracy of predicted sewing connections.	[14], [31], [32], [34]
	Average Matching Distance (AMD)	Matching degree between predicted and GT stitching vertices.	[35]
	Intersection-over-Union (IoU)	Overlap between predicted and GT panels.	[36]
3D Garments	**Chamfer Distance (CD)**	Distance between predicted and GT garment point clouds.	[29], [54]
	Point-to-Surface (P2S)	Distance from predicted points to GT garment surface.	[54]
	Simulation Success Rate	Ratio of patterns that simulate without physical errors.	[28], [29], [54]
	Clothes-to-body Distance	Average distance between the garment and the body model.	[27]
	Aesthetic Score	User rating of visual appeal and coherence.	[28]
Multimodal Reconstruction	**CLIP Score**	Semantic alignment between garment and text prompt.	[29], [30], [35]
	Agreement Score	Expert-rated consistency with the input design prompt.	[28]

Note: GT is an abbreviation for Ground-truth.

a 2D grid and then computing the ratio of the area of their intersection to the area of their union on a pixel-wise basis.

2) Metrics for 3D Garments: In addition to evaluating sewing patterns, assessing the final 3D garment is essential, as it constitutes the ultimate output of the design pipeline. core evaluation criterion is the geometric similarity to the ground-truth garment. This is typically quantified using metrics such as Chamfer Distance (CD), which calculates the average distance between points on the predicted and ground-truth meshes [29], [54]. related measure, the Point-to-Surface (P2S) distance, is used in GarmentX to compute the shortest distance from points on the predicted surface to the ground-truth surface [54].

nother key dimension is physical realism, particularly since garments generated from sewing patterns are intended for simulation and real-world wear. widely used metric in this context is the Simulation Success Rate, which evaluates whether the predicted sewing patterns can be assembled and simulated into physically stable 3D garments without errors [28], [29], [54]. To assess garment-body fit, SewingLDM proposes the Clothes-to-Body Distance, which measures the average distance between the garment mesh and the underlying human body model. Lower values of this metric indicate a better physical fit [27].

Beyond geometric and physical metrics, aesthetic quality and subjective user perception are also important, especially for design-driven applications. These aspects are typically evaluated through user studies in which participants rate the visual and structural quality of 3D garments. For instance, the esthetic Score introduced in Design2GarmentCode captures users' judgments regarding the overall visual appeal and design coherence of the generated garment [28].

3) Metrics for Multimodal Reconstruction: In multimodal reconstruction tasks, evaluation metrics are used to assess the semantic alignment between the generated 3D garment and the original user input. For text-guided generation, the CLIP Score is a widely adopted automated metric that measures the semantic similarity between the rendered 3D garment and the corresponding textual prompt. higher CLIP Score indicates a closer alignment between the visual output and the intended textual description [29], [30], [35].

In addition to automated evaluation, human assessment provides valuable qualitative insights. The greement Score is one such metric, where expert evaluators rate the consistency between the generated garment and the original design input, which may take the form of a text prompt, reference image, or design sketch. This evaluation is essential for determining how well a model interprets and translates multimodal design intent into coherent garment representations [28].

VIII. PPLIC TIONS

. Physical Garment Production

1) Individual Customization: Individual customization generates sewing patterns tailored to a user's ergonomic needs or aesthetic pReferences, offering an efficient alternative to mass production's standardized sizing. This process uses advanced technology to translate user data into manufacturable patterns.

Customization can be guided by two main principles. First, ergonomic customization prioritizes fit and comfort, employing technologies like 3D body scanning for precise geometry capture or automated systems like Sewformer to generate patterns directly from measurements (Figure 9(a)) [6], [23], [32], [46]. Second, aesthetic customization focuses on design, using models trained on fashion datasets to interpret multimodal inputs like images or text. These models infer style preferences and map them to design attributes to create patterns that match the user's desired look (Figure 9(b)) [119], [120], [121], [122].

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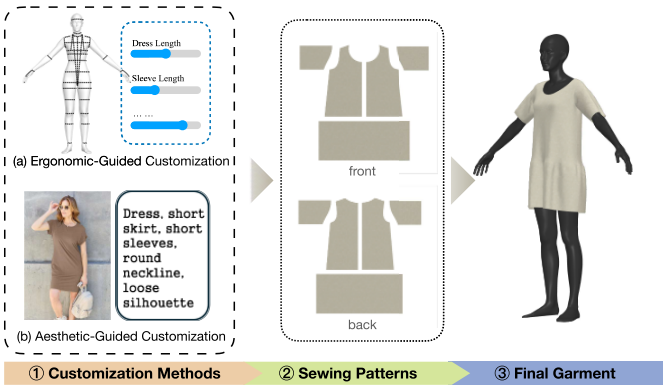


Fig. 9. pplications of individual customization. (a) Ergonomic-Guided Customization, and (b) esthetic-Guided Customization. The pipeline consists of three main steps: (1) customization method is used to (2) generate sewing patterns and (3) visualize them as 3D garments.

2) Mass Customization: Mass customization facilitates the large-scale production of personalized garments by balancing product diversity with manufacturing efficiency. Its key mechanism is size grading, the process of systematically modifying a base sewing pattern to generate a range of standard sizes (e.g., S, M, L) [123], [124], [125]. However, traditional grading methods like uniform scaling can distort garment proportions, especially in complex areas or at size extremes.

To address this challenge, modern approaches use advanced techniques. First, scientific grading systems and digital tools like Gerber and Lectra apply rule-based geometric operations and localized scaling to maintain design integrity across sizes [126], [127], [128]. Second, manufacturers incorporate anthropometric data from target populations to create region-specific standards, as exemplified by UNIQLO's use of sian body data to ensure a better fit [128]. These data-driven methods ensure proportional accuracy and enable more responsive, market-aligned manufacturing.

B. Virtual Garment Simulation

1) Virtual Try-on: In virtual try-on applications, sewing patterns provide the structural foundation for simulating garment fit and movement on 3D human avatars [30], [129], [130]. These patterns enable physics-based simulations that accurately model complex garment behaviors. For instance, platforms such as lter-Ego, CLO3D and Style3D utilize pattern-based models to simulate sleeve stretching during arm movements and fabric compression at contact points with the body [23], [131], [132], [133]. Similarly, systems like ISP handle multi-layer garment interactions by simulating compression between layers, such as between an overcoat and a shirt [48]. Beyond physical simulation, 2D sewing patterns are essential for achieving high visual fidelity. They are applied in UV mapping to accurately project textures, prints, and seam details onto the 3D garment mesh, reducing distortion and enhancing realism [47], [134].

2) Digital Garments: Sewing patterns can also be used in the creation of digital clothing for characters in games and animations [135], [136]. In these virtual environments, garments often encounter more complex challenges than those in standard simulations. They must be adapted to fit stylized or non-humanoid character morphologies, requiring flexible pattern generation methods that accommodate unconventional body shapes [137]. dditionally, the garments must behave realistically under extreme conditions, such as exaggerated character movements involving rapid stretching and rotation, or simulations in unusual physical contexts such as zero gravity. chieving these goals depends on the integration of robust physics-based simulation models with advanced sewing pattern construction techniques, ensuring that the final digital garment exhibits both high visual fidelity and physical plausibility.

IX. Discussions

. Comparisons

To provide a systematic overview of generative fashion design, we compare three major paradigms: 2D fashion design [22], [138], [139], [140], [141], [142], [143], pattern-free 3D fashion design [21], [130], [134], [144], and pattern-driven 3D fashion design [2], [8], [14], [27], [29], [30], [31], [32], [34]. This comparison is particularly relevant given the rapid advances in deep learning and computer graphics that have transformed the field. Our analysis focuses on their core pipelines and key characteristics, including underlying technical approaches, design capabilities, and feasibility for physical production. Figure 10 shows pipelines of 2D fashion design, pattern-free 3D fashion design, and pattern-driven 3D fashion design. Their main distinctions are summarized in Table VIII.

1) Comparison Between 2D Fashion Design and 3D Fashion Design: The primary distinction between 2D and 3D fashion design methods lies in their model architectures, which are dictated by their respective output formats. s summarized in Table VIII, 2D methods employ pixel-based decoders, such as U-Nets and G Ns, to learn mappings for reconstructing 2D garment images via image-to-image translation [18], [19], [20], [143], [147], [148], [150], [172], [173], [174], [175], as shown in Figure 10(a). In contrast, 3D approaches utilize geometry-based decoders to generate structured representations like signed distance fields (SDFs) or point clouds, which explicitly define 3D surface geometry [14], [26], [176], [177], [178], illustrated in Figure 10(b).

The architectural differences between 2D and 3D pipelines define their data requirements and intermediate representations. 2D approaches leverage abundant public image datasets for image-to-image synthesis, typically using segmentation masks as intermediate representations [19], [145], [146], [147], [150], [151], [152], [153], [179]. Conversely, 3D approaches rely on scarce and difficult-to-annotate geometric datasets, employing latent codes as intermediate representations to train [91], [92], [130], [134], [162], [163], [164].

These differences in modeling approaches directly impact downstream applications. Since 2D models operate solely on image pixels, their outputs are limited to surface-level modifications such as texture changes. nd, the output 2D images often only provide a single-viewing perspective, offering limited reference value for production [20], [180], [181],

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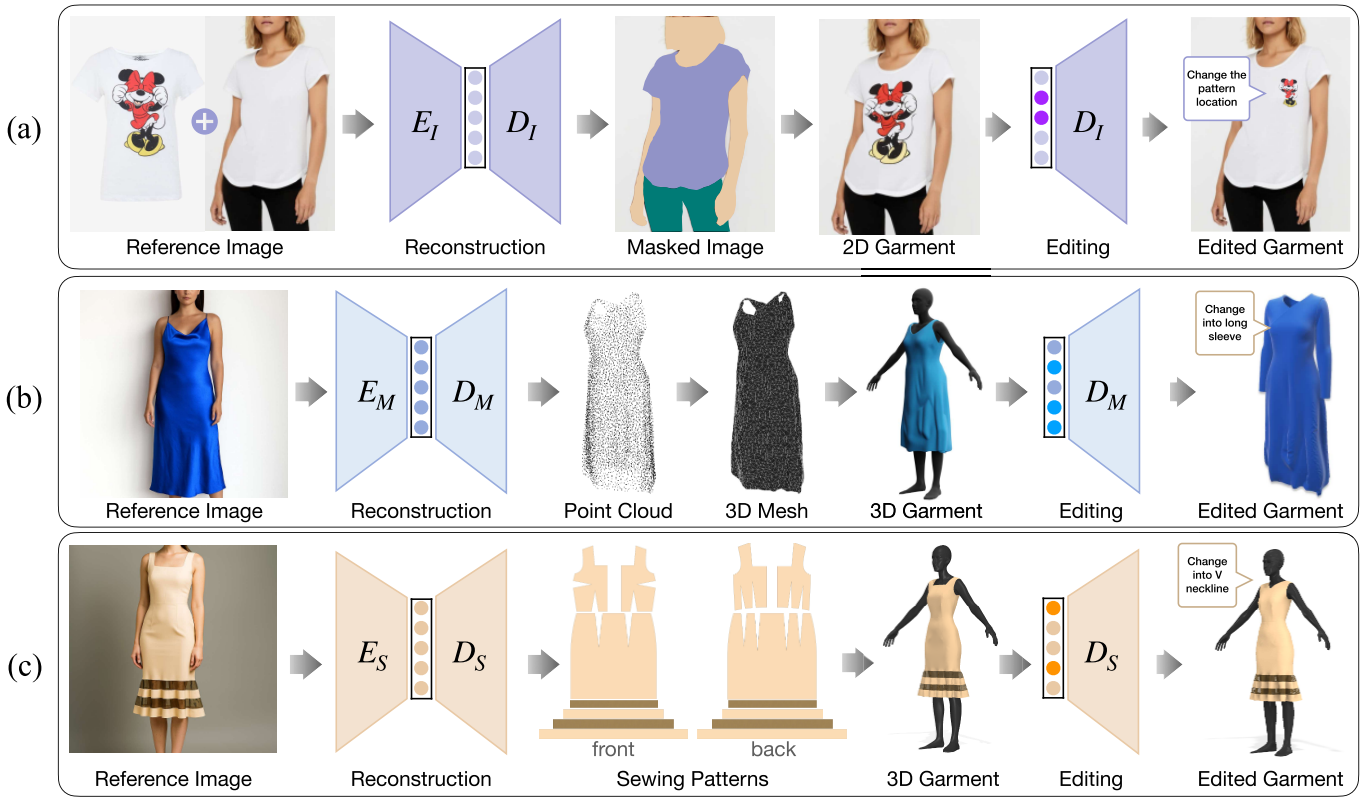


Fig. 10. Comparison of generative pipelines for fashion design, highlighting their core mechanisms. Each pipeline employs an encoder (E) to map the reference image into a latent space and a decoder (D) to generate the output. I for 2D fashion design, M for pattern-free 3D fashion design, and S for pattern-driven 3D fashion design. (a) 2D Fashion Design [18], [19], [145], [146], [147], [148], [149], [150], [151], [152], [153], [154], [155], [156], [157], [158], [159], [160], [161], [172]; (b) Pattern-free 3D Fashion Design [91], [92], [130], [134], [162], [163], [164], [165], [166], [167], [168], [169]; (c) Pattern-driven 3D Fashion Design [2], [3], [5], [8], [14], [28], [29], [30], [31], [32], [33], [34], [36], [44], [48], [51], [54], [170], [171].

T BLE VIII
COMP RISON OF 2D F SHION DESIGN, P TTERN-FREE 3D F SHION DESIGN, ND P TTERN-DRIVEN 3D F SHION DESIGN PPRO CHES

Dimensions	**2D Fashion Design**	**Pattern-free 3D Fashion Design**	**Pattern-driven 3D Fashion Design**
Core Concept	End-to-end 2D clothing image generation	End-to-end 3D geometry generation	Explicitly modeling garment structure, with SP as intermediate representation
Training Dataset	Large-scale image datasets	3D clothing model datasets	Sewing pattern datasets
Encoder Architecture	Multimodal Encoder (CLIP, ViT, ResNet, or LLaMA, etc.)		
Decoder Architecture	Pixel-based generators (U-Net, GAN, or Diffusion models).	Geometry-based generators (SDF, or NeRF).	Structure-based generators (MLP, LSTM, Transformer, or DiT).
Output	2D images (pixel level)	3D models (point clouds, meshes)	2D sewing patterns + 3D models (from pattern stitching simulation)
Creative Freedom	**High** (High visual freedom and less physical constraints)	**High** (Can quickly generate a variety of 3D designs)	**Low** (Limited by the complexity of physical laws and pattern design)
Editability	**Low** (Pixel-level image adjustments)	**Low** (3D model vertex manipulation or latent code editing)	**High** (Editing of parameters to drive 3D model changes)
Manufacturability	**Very Low** (requires manual interpretation and redesign of the sewing patterns)	**Low** (Cannot generate producible panels)	**High** (Output directly usable for cutting production)
Studies	[18]–[20], [145]–[161]	[91], [92], [130], [134], [162]–[169]	[2], [3], [5], [8], [14], [28]–[34], [36], [44], [48], [51], [54], [170], [171]

[182], [183], [184], [185], [186], [187]. In contrast, 3D models produce structured geometric representations that allow for meaningful shape modifications. Furthermore, 3D models can theoretically provide renderings from any viewpoint, making them more valuable for practical production [48], [166], [168], [169], [188], [189], [190], [191], [192]. However, whether 3D models can be directly applied to production will be further discussed in the next part.

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2) Comparison Between Pattern-Free and Pattern-Driven 3D Fashion Design: In 3D fashion design, a critical distinction lies in whether the garment structure is modeled implicitly or explicitly. Pattern-free approaches adopt an end-to-end geometry-based decoder (see Figure 10(b)) to directly predict the final 3D shape, such as a point cloud or mesh [92], [130], [134], [163], [164], [193]. These models learn a direct mapping from the input (e.g., sketches, images, or text) into a complete 3D surface representation. In contrast, pattern-driven approaches employ structure-aware decoders, such as LSTM or Transformer architectures, to generate a set of 2D sewing patterns as an intermediate representation rather than the 3D shape itself (see Figure 10(c)) [1], [3], [8]. This modeling choice leads to a fundamentally different generation process.

The pattern-free pipeline typically follows an end-to-end paradigm, where a single neural model directly outputs the 3D geometry from the input [176], [194], [195], [196], [197]. By contrast, the pattern-driven pipeline is divided into two stages [28], [29], [30], [32]. In the first stage, a neural network generates parametric 2D sewing patterns; in the second stage, these patterns are assembled into a 3D garment through physics-based simulation [55], [60], [62], [198], [199]. The simulation stage relies on canonical principles from cloth simulation and differentiable physics research, which form the foundation for digitally draping 2D panels into 3D forms [200], [201], [202]. This two-stage approach explicitly incorporates physical constraints and manufacturing logic into the design process. Consequently, the generative model produces not meshes, but parametric curves and panel structures that can be directly interpreted and fabricated.

These architectural differences lead to notable trade-offs. Pattern-free methods offer high creative flexibility and can produce highly novel garment shapes. However, their outputs are often difficult to edit and lack direct applicability to manufacturing workflows [135], [136]. In contrast, pattern-driven methods emphasize structured generation, yielding outputs that are inherently parametric, editable, and manufacturable. By treating sewing patterns as the primary design representation, these approaches bridge the gap between digital design and real-world garment production, but the disadvantage is that their design freedom is relatively limited [1], [203], [204].

B. Limitations

lthough the remarkable progress of deep learning-based methods in sewing pattern-driven 3D fashion design, several inherent limitations hinder their widespread adoption in practical applications, particularly concerning data availability and quality.

(1) Deficient Datasets Size and Quality: major limitation is the lack of large-scale, high-quality, and standardized datasets for sewing patterns. Most publicly available datasets are relatively small and lack consistent, richly annotated multimodal information [23], [31]. Moreover, aligning sewing patterns with corresponding 3D models, 2D sketches or images, and textual descriptions is a resource-intensive task that requires significant manual effort and costs. These data bottlenecks directly constrain the performance of multimodal learning models.

(2) Limited Representation Capacity: limitation stems from the data representations currently employed for sewing patterns, which are often too simplistic to capture the full complexity of production-ready garments. Most existing methodologies model a garment as an idealized assembly of 2D panels [2], [25]. This paradigm inherently lacks the capacity to describe crucial, non-panel components such as pockets, plackets, collars, cuffs, or the placement of hardware like zippers and buttons. Existing research has addressed the challenge of enforcing geometric constraints on base panels, such as ensuring stitchability and preventing self-intersections [27]. However, the output sewing patterns still not a truly manufacturable design since the lack of the detailed components [1], [3], [30]. This gap in representation is a major bottleneck preventing the automatic generation of complete and industry-standard sewing patterns.

(3) mbiguous Multimodal Inputs: Reconstructing sewing patterns from high-level inputs poses a big challenge due to the ambiguity and under-specification of source data. Inputs such as a single sketch, a product image, or a brief textual description are often under-specified, they lack the detailed information required to define a single, unique 3D garment [28]. Consequently, a single input can correspond to multiple plausible pattern designs, leading to significant uncertainty and variability in the reconstruction results. The core technical difficulty is to establish a robust mapping between abstract, qualitative information from the input modality and the precise, quantitative parameters of the sewing patterns. For instance, the model must learn how to translate a high-level concept like "loose sleeves" into specific geometric operations, such as widening the sleeve panel and adjusting its boundary curves [30], [31]. However, existing methods resolve this ambiguity not by understanding the abstract principle of "looseness", but by learning a statistical correlation from the data. They generate a single pattern that represents an averaged version of all "loose sleeves" examples in their training set. Effectively learning this cross-modal alignment from abstract concepts to concrete geometry remains a primary open problem in the field.

(4) Unstandardized Evaluation Metrics: nother significant challenge is the lack of a unified evaluation framework. Most studies propose custom metrics tailored to their own models, making it difficult to compare results across different approaches [14], [33]. This absence of standardization prevents the field from establishing reliable benchmarks for technical performance. Furthermore, the evaluation of complex, qualitative attributes like design quality and aesthetics remains superficial. Most studies rely on subjective user ratings or expert assessments, which, while valuable, suffer from low scalability and potential bias, failing to provide deep insights into why a design is considered successful [28], [29]. Consequently, there is an urgent need to develop both a standardized suite of metrics for core technical performance and more sophisticated, objective methods for assessing the crucial aspects of garment design.

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C. Future Work

We identify four key research directions for future research that can advance the field of sewing pattern-driven 3D fashion design.

(1) Building Large-Scale nnotated Datasets: primary focus should be on addressing the current scarcity and quality issues of sewing pattern datasets. Future efforts must extend beyond simply increasing dataset size to curating large-scale, richly annotated benchmarks. These datasets should include fine-grained manufacturing details such as darts, pleats, seam allowances, and corresponding fabric properties, which are often missing. To facilitate this, developing semi-automated annotation pipelines is crucial for efficiently generating precisely aligned data tuples that link sewing patterns to their corresponding 3D models, 2D images, and textual descriptions. Furthermore, exploring procedural generation techniques and specialized data augmentation strategies tailored for pattern geometry could effectively synthesize diverse, high-quality samples, thereby mitigating the data bottleneck for training more capable models.

(2) Developing Holistic and Structured Representations: To transition from idealized digital models to precise sewing patterns, future research must prioritize the fundamental enhancement of sewing pattern representations. The panel-centric paradigm needs to evolve into a more holistic, structured framework that cover the full complexity of a garment. This requires establishing a richer descriptive vocabulary that formalizes not only base panels but also ancillary components such as pockets, collars, and plackets as distinct parametric entities. Moreover, this representation must integrate specifications for non-fabric hardware, detailing the precise placement and type of zippers and buttons, while also encoding the intricate assembly logic that dictates the interconnection of all components. This would enable learning-based systems to directly generate complete, manufacturable patterns.

(3) Deepening Multimodal Understanding: ddressing the challenges in multimodal reconstruction requires developing more robust models for handling ambiguous, high-level inputs. Future research should investigate probabilistic generative frameworks, such as conditional variational autoencoders (CV Es), that can produce a distribution of plausible pattern solutions from a single under-specified input like a sketch or text. This approach would better reflect the inherent one-to-many nature of the design process. Furthermore, an interactive pipeline where users can provide iterative feedback to disambiguate an initial prompt would be a valuable direction, allowing for the progressive refinement of generated patterns to more accurately match a designer's specific intent.

(4) Formulating Standardized Evaluation Metrics: To foster systematic progress, the field requires a standardized evaluation framework that assesses the intrinsic quality and manufacturability of the sewing patterns. For instance, new metrics should assess a pattern's core manufacturability by quantifying its geometric validity (e.g., checking for self-intersections), stitchability (e.g., measuring seam length discrepancies), and stitching correctness (e.g., verifying the appropriate panel set). For assessing qualitative attributes like aesthetic appeal and fit, research should aim to create objective, automated metrics that can supplement subjective user studies. For example, models could be trained to predict aesthetic scores, while garment fit could be evaluated using simulation-based metrics like body-garment distance or pressure distribution maps.

X. Conclusion

This survey has presented a systematic review of deep learning for sewing pattern-driven 3D fashion design. We introduced a novel four-stage pipeline encompassing Representation, Generation, Reconstruction, and Editing to structure the field and chart its technological progression. Our analysis traced the evolution from foundational geometric encoding techniques to sophisticated deep learning models capable of synthesizing and modifying complex garment designs. We specifically highlighted the crucial task of reconstructing manufacturable patterns from multimodal inputs like images and text, and the emerging trend of intuitive, language-driven editing, which lowers the barrier to digital design.

Despite these advancements, our review identifies several persistent challenges that hinder practical application. primary bottleneck is the scarcity of large-scale, high-quality datasets featuring standardized, fine-grained annotations for production-ready patterns. nother significant limitation lies in clothing representation, as existing methods struggle to accurately model complex garments. Furthermore, in multimodal parsing, the fashion-specific semantic understanding of current models remains insufficient. Finally, the field lacks standardized evaluation metrics; current assessments often prioritize geometric similarity while neglecting crucial aspects such as aesthetics, reconstruction fidelity, and fit.

Future research should prioritize the development of comprehensive data ecosystems and the design of novel representation and deep learning architectures capable of reasoning about geometric and physical constraints. Concurrently, advancing multimodal models and establishing robust evaluation metrics is essential for resolving input ambiguity. Ultimately, these efforts will enable the creation of intelligent systems that can reliably construct generative sewing patterns suitable for real-world manufacturing.

References

[1] N. Pietroni, C. Dumery, R. Falque, M. Liu, T. Vidal-Calleja, and O. Sorkine-Hornung, "Computational pattern making from 3D garment models," *CM Trans. Graph.*, vol. 41, no. 4, pp. 1–13, Jul. 2022.

[2] M. Korosteleva and S. Lee, "Generating datasets of 3D garments with sewing patterns," in *Proc. Neural Inf. Process. Syst. Track Datasets Benchmarks*, 2021, pp. 1–10.

[3] F. Berthouzoz, . Garg, D. M. Kaufman, E. Grinspun, and M. grawala, "Parsing sewing patterns into 3D garments," *CM Trans. Graph.*, vol. 32, no. 4, pp. 1–12, Jul. 2013.

[4] M. Jankoska, " pplication C D methods in 3D clothing design," *Tekstilna Industrija*, vol. 68, no. 4, pp. 31–37, 2020.

[5] K. Wolff and O. Sorkine-Hornung, "Wallpaper pattern alignment along garment seams," *CM Trans. Graph.*, vol. 38, no. 4, pp. 1–12, 2019.

[6] H. Wang, "Rule-free sewing pattern adjustment with precision and efficiency," *CM Trans. Graph.*, vol. 37, no. 4, pp. 1–13, ug. 2018.

[7] E. Scott and . S. M. Sayem, "Landmarking and measuring for critical body shape analysis targeting garment fit," in *Proc. Int. Conf. Exhib. 3D Body Scanning Process. Technol.*, 2018, pp. 222–235.

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[8] M. Korosteleva and O. Sorkine-Hornung, "GarmentCode: Programming parametric sewing patterns," *CM Trans. Graph.*, vol. 42, no. 6, pp. 1–15, Dec. 2023.

[9] K. Liu et al., "3D interactive garment pattern-making technology," *Comput.- ided Design*, vol. 104, pp. 113–124, Nov. 2018.

[10] Y. Meng, P. Y. Mok, and X. Jin, "Computer aided clothing pattern design with 3D editing and pattern alteration," *Comput.- ided Design*, vol. 44, no. 8, pp. 721–734, ug. 2012.

[11] M. . Habib and M. S. lam, " comparative study of 3D virtual pattern and traditional pattern making," *J. Textile Sci. Technol.*, vol. 10, no. 1, pp. 1–24, 2024.

[12] K.-H. Choi, "3D dynamic fashion design development using digital technology and its potential in online platforms," *Fashion Textiles*, vol. 9, no. 1, p. 9, Mar. 2022.

[13] S. Lu, P. Y. Mok, and X. Jin, " new design concept: 3D to 2D textile pattern design for garments," *Comput.- ided Design*, vol. 89, pp. 35–49, ug. 2017.

[14] M. Korosteleva and S.-H. Lee, "NeuralTailor: Reconstructing sewing pattern structures from 3D point clouds of garments," *CM Trans. Graph.*, vol. 41, no. 4, pp. 1–16, Jul. 2022.

[15] . Imtiaz, N. Pathirana, S. Saheel, K. Karunanayaka, and C. Trenado, " review on the influence of deep learning and generative I in the fashion industry," *J. Future rtif. Intell. Technol.*, vol. 1, no. 3, pp. 201–216, Oct. 2024.

[16] H. Sadr et al., "Unveiling the potential of artificial intelligence in revolutionizing disease diagnosis and prediction: comprehensive review of machine learning and deep learning approaches," *Eur. J. Med. Res.*, vol. 30, no. 1, p. 418, May 2025.

[17] D. Simian and F. Husac, "Challenges and opportunities in deep learning driven fashion design and textiles patterns development," in *Modelling and Development of Intelligent Systems*. Cham, Switzerland: Springer, 2023, pp. 173–187.

[18] Y. R. Cui, Q. Liu, C. Y. Gao, and Z. Su, "FashionG N: Display your fashion design using conditional generative adversarial nets," *Comput. Graph. Forum*, vol. 37, no. 7, pp. 109–119, Oct. 2018.

[19] X. Han, Z. Wu, Z. Wu, R. Yu, and L. S. Davis, "VITON: n image-based virtual try-on network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7543–7552.

[20] D. Song, J.-H. Zeng, M. Liu, X.-Y. Li, and .- . Liu, "Fashion customization: Image generation based on editing clue," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 6, pp. 4434–4444, Jun. 2024.

[21] N. Sarafianos, T. Stuyck, X. Xiang, Y. Li, J. Popovic, and R. Ranjan, "Garment3DGen: 3D garment stylization and texture generation," in *Proc. Int. Conf. 3D Vis. (3DV)*, Mar. 2025, pp. 1–18.

[22] Y. Xu, T. Gu, W. Chen, and C. Chen, "OOTDiffusion: Outfitting fusion based latent diffusion for controllable virtual try-on," 2024, *arXiv:2403.01779*.

[23] M. Korosteleva et al., "GarmentCodeData: dataset of 3D made-to-measure garments with sewing patterns," in *Proc. Eur. Conf. Comput. Vis.*, 2024, pp. 110–127.

[24] S. Lim, S. Kim, and S.-H. Lee, "SPnet: Estimating garment sewing patterns from a single image of a posed user," in *Proc. Eurographics*, 2024, pp. 1–9.

[25] T. Y. Wang, D. Ceylan, J. Popović, and N. J. Mitra, "Learning a shared shape space for multimodal garment design," *CM Trans. Graph.*, vol. 37, no. 6, pp. 1–13, Dec. 2018.

[26] S. Nag, . Qi, X. Zhu, and . Shamir, "PersonalTailor: Personalizing 2D pattern design from 3D garment point clouds," 2023, *arXiv:2303.09695*.

[27] S. Liu et al., "Multimodal latent diffusion model for complex sewing pattern generation," 2024, *arXiv:2412.14453*.

[28] F. Zhou et al., "Design2GarmentCode: Turning design concepts to tangible garments through program synthesis," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 23712–23722.

[29] S. Bian et al., "ChatGarment: Garment estimation, generation and editing via large language models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 2924–2934.

[30] K. He, K. Yao, Q. Zhang, J. Yu, L. Liu, and L. Xu, "DressCode: utoregressively sewing and generating garments from text guidance," *CM Trans. Graph.*, vol. 43, no. 4, pp. 1–13, Jul. 2024.

[31] K. Nakayama et al., " Ipparel: multimodal foundation model for digital garments," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 8138–8149.

[32] L. Liu, X. Xu, Z. Lin, J. Liang, and S. Yan, "Towards garment sewing pattern reconstruction from a single image," *CM Trans. Graph.*, vol. 42, no. 6, pp. 1–15, Dec. 2023.

[33] C.-H. Chen, J.-W. Su, M.-C. Hu, C.-Y. Yao, and H.-K. Chu, "Panelformer: Sewing pattern reconstruction from 2D garment images," in *Proc. IEEE/CVF Winter Conf. ppl. Comput. Vis. (W CV)*, Jan. 2024, pp. 443–452.

[34] X. Li, Q. Yao, and Y. Wang, "GarmentDiffusion: 3D garment sewing pattern generation with multimodal diffusion transformers," 2025, *arXiv:2504.21476*.

[35] S. Li et al., "GarmageNet: multimodal generative framework for sewing pattern design and generic garment modeling," 2025, *arXiv:2504.01483*.

[36] Y. Tatsukawa, . Qi, I.-C. Shen, and T. Igarashi, "GarmentImage: Raster encoding of garment sewing patterns with diverse topologies," 2025, *arXiv:2505.02592*.

[37] W. Shi, W. Wong, and X. Zou, "Generative I in fashion: Overview," *CM Trans. Intell. Syst. Technol.*, vol. 16, Feb. 2025, rt. no. 3718098.

[38] Y. Xu, S. Thomassey, and X. Zeng, " I for apparel manufacturing in big data era: focus on cutting and sewing," in *rtificial Intelligence for Fashion Industry in the Big Data Era*. Singapore: Springer, 2018, pp. 125–151.

[39] Y. Sun, Z. Hao, Z. Wang, J. Jin, Q. Ye, and Y. Lyu, "Deep learning for 3D garment generation: review," *Textile Res. J.*, vol. 95, nos. 23–24, pp. 2882–2908, Dec. 2025.

[40] X. Chen, G. Wang, D. Zhu, X. Liang, P. H. S. Torr, and L. Lin, "Structure-preserving 3D garment modeling with neural sewing machines," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2022, pp. 1–13.

[41] Y. Shen, J. Liang, and M. C. Lin, "G N-based garment generation using sewing pattern images," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 225–247.

[42] K. Wolff, P. Herholz, and O. Sorkine-Hornung, "Reflection symmetry in textured sewing patterns," in *Proc. Symp. Vis., Modeling Visualizat.*, 2019, pp. 1–8.

[43] C. Liu, W. Xu, Y. Yang, and H. Wang, " utomatic digital garment initialization from sewing patterns," *CM Trans. Graph.*, vol. 43, no. 4, pp. 1–12, Jul. 2024.

[44] Y. Zhang, L. Ma, Z. Guo, T. Li, and F. Zou, "Personalized garment pattern generation based on space vector and distance ease," *Int. J. Clothing Sci. Technol.*, vol. 35, no. 5, pp. 715–737, Sep. 2023.

[45] C. Curtis, " document format for sewing patterns," in *Proc. CM Symp. Document Eng.*, ug. 2023, pp. 1–4.

[46] H.-Y. Chen et al., "Dress anyone: utomatic physically-based garment pattern refitting," 2024, *arXiv:2405.19148*.

[47] R. Li, C. Cao, C. Dumery, Y. You, H. Li, and P. Fua, "Single view garment reconstruction using diffusion mapping via pattern coordinates," 2025, *arXiv:2504.08353*.

[48] R. Li, B. Guillard, and P. Fua, "ISP: Multi-layered garment draping with implicit sewing patterns," in *Proc. dv. Neural Inf. Process. Syst.*, 2023, pp. 1–12.

[49] T. Igarashi, T. Moscovich, and J. F. Hughes, " s-rigid-as-possible shape manipulation," *CM Trans. Graph.*, vol. 24, no. 3, pp. 1134–1141, Jul. 2005.

[50] . Bartle, . Sheffer, V. G. Kim, D. M. Kaufman, N. Vining, and F. Berthouzoz, "Physics-driven pattern adjustment for direct 3D garment editing," *CM Trans. Graph.*, vol. 35, no. 4, pp. 1–11, Jul. 2016.

[51] P. Jin et al., "Design and research of automatic Garment-Pattern-Generation system based on parameterized design," *Sustainability*, vol. 15, no. 2, p. 1268, Jan. 2023.

[52] S. Krzywinski and J. Siegmund, "3D product development for loose-fitting garments based on parametric human models," *IOP Conf. Ser., Mater. Sci. Eng.*, vol. 254, Oct. 2017, rt. no. 152006.

[53] S. Yang et al., "Physics-inspired garment recovery from a single-view image," *CM Trans. Graph.*, vol. 37, no. 5, pp. 1–14, Oct. 2018.

[54] J. Guo et al., "GarmentX: utoregressive parametric representations for high-fidelity 3D garment generation," 2025, *arXiv:2504.20409*.

[55] X. Li et al., "Dress-1-to-3: Single image to simulation-ready 3D outfit with diffusion prior and differentiable physics," 2025, *arXiv:2502.03449*.

[56] M. Shi, Y. Wei, L. Chen, D. Zhu, T. Mao, and Z. Wang, "Learning a shared deformation space for efficient design-preserving garment transfer," *Graph. Models*, vol. 115, May 2021, rt. no. 101106.

[57] C. Bao, Y. Miao, B. Gu, K. Liu, and Z. Liu, "3D interactive garment parametric pattern-making and linkage editing based on constrained contour lines," *Int. J. Clothing Sci. Technol.*, vol. 33, no. 5, pp. 696–723, Nov. 2021.

[58] Z. Su et al., "MulayCap: Multi-layer human performance capture using a monocular video camera," *IEEE Trans. Vis. Comput. Graph.*, vol. 28, no. 4, pp. 1862–1879, pr. 2022.

Authorized licensed use limited to: Netaji Subhas University of Technology New Delhi. Downloaded on August 31,2026 at 07:56:21 UTC from IEEE Xplore. Restrictions apply.

[59] N. Hasler, B. Rosenhahn, and H.-P. Seidel, "Reverse engineering garments," in *Computer Vision/Computer Graphics Collaboration Techniques*. Berlin, Germany: Springer, 2007, pp. 200–211.

[60] B. Yu, F. Cordier, and H. Seo, "Inverse garment and pattern modeling with a differentiable simulator," *Comput. Graph. Forum*, vol. 43, no. 7, p. 15249, Oct. 2024.

[61] W. H. Press, W. T. Vetterling, S. . Teukolsky, and B. P. Flannery, *Numerical Recipes in C+: The rt of Scientific Computing*, 2nd ed., Cambridge, U.K.: Cambridge Univ. Press, 2001, pp. 1–1002.

[62] J. Liang, M. C. Lin, and V. Koltun, "Differentiable cloth simulation for inverse problems," in *Proc. dv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 771–780.

[63] R. Narain, . Samii, and J. F. O'Brien, " daptive anisotropic remeshing for cloth simulation," *CM Trans. Graph.*, vol. 31, no. 6, pp. 1–10, Nov. 2012.

[64] M. Mernik, J. Heering, and . M. Sloane, "When and how to develop domain-specific languages," *CM Comput. Surv.*, vol. 37, no. 4, pp. 316–344, Dec. 2005.

[65] Z. Wang, X. Tao, X. Zeng, Y. Xing, Z. Xu, and P. Bruniaux, "Design of customized garments towards sustainable fashion using 3D digital simulation and machine learning-supported human–product interactions," *Int. J. Comput. Intell. Syst.*, vol. 16, no. 1, p. 16, Feb. 2023.

[66] M. Nazari, H. Sadr, H. Emami, and R. Rabiei, " telemonitoring system for high-risk pregnancies: novel approach to enhancing prenatal care and reducing maternal and fetal complications," in *Proc. 11th Int. Conf. Web Res. (ICWR)*, pr. 2025, pp. 431–437.

[67] C. Nash, Y. Ganin, S. M. . Eslami, and P. Battaglia, "PolyGen: n autoregressive generative model of 3D meshes," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 7220–7229.

[68] Y. Siddiqui et al., "MeshGPT: Generating triangle meshes with decoder-only transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 19615–19625.

[69] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, Nov. 1997.

[70] . Vaswani et al., " ttention is all you need," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, vol. 30, 2025, pp. 5998–6008.

[71] . Mao, M. Mohri, and Y. Zhong, "Cross-entropy loss functions: Theoretical analysis and applications," in *Proc. Int. Conf. Mach. Learn.*, 2023, pp. 1–26.

[72] . Dosovitskiy et al., " n image is worth 16 16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[73] R. Rombach, . Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 10684–10695.

[74] C. Du, J. Wang, K. Liu, and S. Xiong, "GLV: Geometric correlation distillation for latent diffusion-enhanced parser-free virtual try-on," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 9, pp. 9175–9189, Sep. 2025.

[75] . Nichol and P. Dhariwal, "Improved denoising diffusion probabilistic models," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 8162–8171.

[76] W. Peebles and S. Xie, "Scalable diffusion models with transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 4172–4182.

[77] J. S. Chen et al., "Pix rt- : Fast training of diffusion transformer for photorealistic text-to-image synthesis," in *Proc. 12th Int. Conf. Learn. Represent.*, 2023, pp. 1–31.

[78] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," 2020, *arXiv:2010.02502*.

[79] Y. Dang, T. Ruan Wan, L. Xi, and W. Tang, " computer-vision based framework for virtual 3D garment reconstruction," *Multimedia Tools ppl.*, vol. 84, no. 24, pp. 27963–27985, Sep. 2024.

[80] C. C. L. Wang, Y. Wang, and M. M. F. Yuen, "Feature based 3D garment design through 2D sketches," *Comput.- ided Design*, vol. 35, no. 7, pp. 659–672, Jun. 2003.

[81] G. Tsai and M. C. Yang, "How it is made matters: Distinguishing traits of designs created by sketches, prototypes, and C D," in *Proc. 29th Int. Conf. Design Theory Methodol.*, ug. 2017, pp. 1–12.

[82] . Fondevilla, D. Rohmer, S. Hahmann, . Bousseau, and M.-P. Cani, "Fashion transfer: Dressing 3D characters from stylized fashion sketches," *Comput. Graph. Forum*, vol. 40, no. 6, pp. 466–483, Sep. 2021.

[83] . Jain, D. Modi, R. Jikadra, and S. Chachra, "Text to image generation of fashion clothing," in *Proc. 6th Int. Conf. Comput. Sustain. Global Develop. (INDI Com)*, Mar. 2019, pp. 355–358.

[84] M. Pernuš, C. Fookes, V. Štruc, and S. Dobrišek, "FICE: Text-conditioned fashion-image editing with guided G N inversion," *Pattern Recognit.*, vol. 158, pp. 1–18, Feb. 2025.

[85] . Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 8748–8763.

[86] M. Nazari, H. Sadr, S. Rostami, and Z. Khodaverdian, "Enhancing web page classification through a semantic-aware and efficient focused crawling methodology," in *Proc. 11th Int. Conf. Web Res. (ICWR)*, pr. 2025, pp. 169–174.

[87] C. Raffel et al., "Exploring the limits of transfer learning with a unified text-to-text transformer," *J. Mach. Learn. Res.*, vol. 21, no. 140, pp. 1–67, 2019.

[88] H. Liu, C. Li, Q. Wu, and Y. J. Lee, "Visual instruction tuning," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2023, pp. 34892–34916.

[89] H. Touvron et al., "LLaM : Open and efficient foundation language models," 2023, *arXiv:2302.13971*.

[90] V. Zouhar et al., " formal perspective on byte-pair encoding," in *Proc. Findings ssoc. Comput. Linguistics, CL*, 2023, pp. 598–614.

[91] Z. Qi et al., "Tailor3D: Customized 3D assets editing and generation with dual-side images," 2024, *arXiv:2407.06191*.

[92] C. Zhang et al., "FabricDiffusion: High-fidelity texture transfer for 3D garments generation from in-the-wild images," in *Proc. SIGGR PH sia Conf. Papers*, Dec. 2024, pp. 1–13.

[93] X. Li, J. Zhang, Y.-K. Lai, J. Yang, and K. Li, "High-quality animatable dynamic garment reconstruction from monocular videos," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 6, pp. 4243–4256, Jun. 2024.

[94] V. Parekh, K. Shaik, S. Biswas, and M. Chelliah, "Fine-grained visual attribute extraction from fashion wear," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2021, pp. 3968–3972.

[95] V. Parekh, S. Mathur, S. Biswas, and K. Shaik, "VPER: Visual product entity recognition for fashion-wear," in *Proc. 5th Joint Int. Conf. Data Sci. Manage. Data (9th CM IKDD CODS 27th COM D)*, Jan. 2022, pp. 270–274.

[96] U. S. Malhi, J. Zhou, C. Yan, . Rasool, S. Siddeeq, and M. Du, "Unsupervised deep embedded clustering for high-dimensional visual features of fashion images," *ppl. Sci.*, vol. 13, no. 5, p. 2828, Feb. 2023.

[97] V. Badrinarayanan, . Kendall, and R. Cipolla, "SegNet: deep convolutional encoder–decoder architecture for image segmentation," *IEEE Trans. Pattern nal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017.

[98] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.- ssist. Intervent.*, 2015, pp. 234–241.

[99] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[100] H. Sadr, M. Nazari, S. Yousefzadeh-Chabok, H. Emami, R. Rabiei, and . shraf, "Enhancing brain tumor classification in MRI images: deep learning-based approach for accurate diagnosis," *Image Vis. Comput.*, vol. 159, Jun. 2025, rt. no. 105555.

[101] F. Mohades Deilami, H. Sadr, and M. Nazari, "Using machine learning-based models for personality recognition," *Big Data Comput. Vis.*, vol. 1, no. 3, pp. 128–139, 2021.

[102] H. Sadr, Z. Khodaverdian, M. Nazari, and M. Yamaghani, " shallow convolutional neural network for cerebral neoplasm detection from magnetic resonance imaging," *Big Data Comput. Vis.*, vol. 4, no. 2, pp. 95–109, 2024.

[103] X. Zhai, B. Mustafa, . Kolesnikov, and L. Beyer, "Sigmoid loss for language image pre-training," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 11941–11952.

[104] M. Oquab et al., "DINOv2: Learning robust visual features without supervision," 2023, *arXiv:2304.07193*.

[105] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 15979–15988.

[106] E. Turquin, J. Wither, L. Boissieux, M.-P. Cani, and J. Hughes, " sketch-based interface for clothing virtual characters," *IEEE Comput. Graph. ppl.*, vol. 27, no. 1, pp. 72–81, Jan. 2007.

[107] P. Y. Mok, J. Xu, X. X. Wang, J. T. Fan, Y. L. Kwok, and J. H. Xin, " n IG -based design support system for realistic and practical fashion designs," *Comput.- ided Design*, vol. 45, no. 11, pp. 1442–1458, Nov. 2013.

[108] K. Liu, X. Zeng, X. Tao, and P. Bruniaux, " ssociate design of fashion sketch and pattern," *IEEE ccess*, vol. 7, pp. 48830–48837, 2019.

Authorized licensed use limited to: Netaji Subhas University of Technology New Delhi. Downloaded on August 31,2026 at 07:56:21 UTC from IEEE Xplore. Restrictions apply.

[109] H. Yan et al., "Toward intelligent design: n I-based fashion designer using generative adversarial networks aided by sketch and rendering generators," *IEEE Trans. Multimedia*, vol. 25, pp. 2323–2338, 2023.

[110] D. Zhang, Y. Liu, J. Wang, and J. Li, " n integrated method of 3D garment design," *J. Textile Inst.*, vol. 109, no. 12, pp. 1595–1605, Dec. 2018.

[111] Y. Zhang, T. Zhang, and H. Xie, "TexControl: Sketch-based two-stage fashion image generation using diffusion model," in *Proc. Nicograph Int. (NicoInt)*, Jun. 2024, pp. 64–68.

[112] D. Ha and D. Eck, " neural representation of sketch drawings," 2017, *arXiv:1704.03477*.

[113] Z. Su et al., "Pixel difference networks for efficient edge detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 5097–5107.

[114] N. Umetani, D. M. Kaufman, T. Igarashi, and E. Grinspun, "Sensitive couture for interactive garment modeling and editing," in *Proc. CM SIGGR PH Papers*, Jul. 2011, pp. 1–11.

[115] . Qi and T. Igarashi, "PerfectTailor: Scale-preserving 2-D pattern adjustment driven by 3-D garment editing," *IEEE Comput. Graph. ppl.*, vol. 44, no. 4, pp. 126–132, Jul. 2024.

[116] P. Volino, N. Magnenat-Thalmann, and F. Faure, " simple approach to nonlinear tensile stiffness for accurate cloth simulation," *CM Trans. Graph.*, vol. 28, no. 4, pp. 1–16, ug. 2009.

[117] H. Bertiche, M. Madadi, and S. Escalera, "CLOTH3D: Clothed 3D humans," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 344–359.

[118] B. Bhatnagar, G. Tiwari, C. Theobalt, and G. Pons-Moll, "Multi-garment Net: Learning to dress 3D people from images," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 5419–5429.

[119] W. Yu, H. Zhang, X. He, X. Chen, L. Xiong, and Z. Qin, " esthetic-based clothing recommendation," in *Proc. World Wide Web Conf. (WWW)*, 2018, pp. 649–658.

[120] Y. Ding, Y. Ma, W. Fan, Y. Yao, T.-S. Chua, and Q. Li, "FashionReGen: LLM-empowered fashion report generation," in *Proc. CM Web Conf. Companion*, May 2024, pp. 991–994.

[121] P. J. Chia et al., "Contrastive language and vision learning of general fashion concepts," *Sci. Rep.*, vol. 12, no. 1, p. 18958, Nov. 2022.

[122] S. Mirchandani et al., "FaD-VLP: Fashion vision-and-language pre-training towards unified retrieval and captioning," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2022, pp. 10484–10497.

[123] N. . Schofield and K. L. LaBat, "Exploring the relationships of grading, sizing, and anthropometric data," *Clothing Textiles Res. J.*, vol. 23, no. 1, pp. 13–27, Jan. 2005.

[124] Y. W. Null and H. Huang, "Three dimensional pattern grading based on deformable body features and 3D developable surface," *J. Fiber Bioeng. Informat.*, vol. 4, no. 2, pp. 115–128, Jun. 2011.

[125] T. Peng, Y. Wang, L. Li, J. Qin, J. Liu, and X. Hu, "VariSize: dataset and model for 3D garment grading and editing based on latent space," in *Proc. IEEE Int. Symp. Product Compliance Eng.- sia (ISPCE- SI)*, Oct. 2024, pp. 1–6.

[126] E. Bye, K. LaBat, E. McKinney, and D.-E. Kim, "Optimized pattern grading," *Int. J. Clothing Sci. Technol.*, vol. 20, no. 2, pp. 79–92, Feb. 2008.

[127] R. Brouet, . Sheffer, L. Boissieux, and M.-P. Cani, "Design preserving garment transfer," *CM Trans. Graph.*, vol. 31, no. 4, pp. 1–11, Jul. 2012.

[128] M. Rosca, .-D. Vatra, and M. vadanei, "The digital transformation of garment product development," *Industria Textila*, vol. 74, no. 1, pp. 98–106, Feb. 2023.

[129] Y. Kwon, S. Petrangeli, D. Kim, H. Wang, V. Swaminathan, and H. Fuchs, "Tailor me: n editing network for fashion attribute shape manipulation," in *Proc. IEEE/CVF Winter Conf. ppl. Comput. Vis. (W CV)*, Jan. 2022, pp. 3142–3151.

[130] . Srivastava, P. Manu, . Raj, V. Jampani, and . Sharma, "WordRobe: Text-guided generation of textured 3D garments," in *Proc. Eur. Conf. Comput. Vis.*, 2025, pp. 458–475.

[131] S.-K. Wong, "Modeling and simulation techniques for garments," in *Computer Technology for Textiles and pparel*. Sawston, C , US : Woodhead Publishing, 2011, pp. 173–199.

[132] K. Wolff, P. Herholz, V. Ziegler, F. Link, N. Brügel, and O. Sorkine-Hornung, "Designing personalized garments with body movement," *Comput. Graph. Forum*, vol. 42, no. 1, pp. 180–194, Feb. 2023.

[133] D. Song et al., "Better fit: ccommodate variations in clothing types for virtual try-on," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 5, pp. 4287–4299, May 2025.

[134] B. Li et al., "GarmentDreamer: 3DGS guided garment synthesis with diverse geometry and texture details," 2024, *arXiv:2405.12420*.

[135] O. Nylén, P. Pall, Y. Ishiwaka, K. Suda, and M. Fratarcangeli, "Interactive assembly and animation of 3D digital garments," in *Proc. Eurographics*, 2020, pp. 73–76.

[136] J. Leaf, D. Sebastian Minor, G. Daviet, N. Chentanez, G. Klar, and E. Quigley, "Bolt: Clothing virtual characters at scale," 2025, *arXiv:2504.17614*.

[137] Z. Sun et al., "Tailor: n integrated text-driven CG-ready human and garment generation system," 2025, *arXiv:2503.12052*.

[138] H.-J. Chen, H.-H. Shuai, and W.-H. Cheng, " survey of artificial intelligence in fashion," *IEEE Signal Process. Mag.*, vol. 40, no. 3, pp. 64–73, May 2023.

[139] Y. Choi, S. Kwak, K. Lee, H. Choi, and J. Shin, "Improving diffusion models for authentic virtual try-on in the wild," in *Proc. Eur. Conf. Comput. Vis.*, 2024, pp. 206–235.

[140] J. Gou, S. Sun, J. Zhang, J. Si, C. Qian, and L. Zhang, "Taming the power of diffusion models for high-quality virtual try-on with appearance flow," in *Proc. 31st CM Int. Conf. Multimedia*, 2023, pp. 7599–7607.

[141] R. Wang et al., "StableGarment: Garment-centric generation via stable diffusion," 2024, *arXiv:2403.10783*.

[142] . Lampe, J. Stopar, D. K. Jain, S. Omachi, P. Peer, and V. Štruc, "DiCTI: Diffusion-based clothing designer via text-guided input," in *Proc. IEEE 18th Int. Conf. utom. Face Gesture Recognit. (FG)*, May 2024, pp. 1–9.

[143] Z. Guo, Z. Zhu, Y. Li, S. Cao, H. Chen, and G. Wang, " I assisted fashion design: review," *IEEE ccess*, vol. 11, pp. 88403–88415, 2023.

[144] X. Sun, Z. Zhang, Y. Tai, H. Tang, Z. Yi, and J. Yang, "Barbie: Text to barbie-style 3D avatars," 2024, *arXiv:2408.09126*.

[145] S. Cao, W. Chai, S. Hao, and G. Wang, "Image reference-guided fashion design with structure-aware transfer by diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2023, pp. 3525–3529.

[146] G. Cazenavette, T. Wang, . Torralba, . . Efros, and J.-Y. Zhu, "Wearable ImageNet: Synthesizing tileable textures via dataset distillation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2022, pp. 2277–2281.

[147] B. Wang, H. Zheng, X. Liang, Y. Chen, L. Lin, and M. Yang, "Toward characteristic-preserving image-based virtual try-on network," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 1–16.

[148] R. Yu, X. Wang, and X. Xie, "VTNFP: n image-based virtual try-on network with body and clothing feature preservation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 10510–10519.

[149] S. Jiang, J. Li, and Y. Fu, "Deep learning for fashion style generation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 9, pp. 4538–4550, Sep. 2022.

[150] S. Cao, W. Chai, S. Hao, Y. Zhang, H. Chen, and G. Wang, "DiffFashion: Reference-based fashion design with structure-aware transfer by diffusion models," *IEEE Trans. Multimedia*, vol. 26, pp. 3962–3975, 2024.

[151] F. Liao, X. Zou, and W. Wong, "Uni-DlLoR : Style fine-tuning for fashion image translation," in *Proc. 32nd CM Int. Conf. Multimedia*, Oct. 2024, pp. 6404–6413.

[152] D. Morelli, . Baldrati, G. Cartella, M. Cornia, M. Bertini, and R. Cucchiara, "LaDI-VTON: Latent diffusion textual-inversion enhanced virtual try-on," in *Proc. 31st CM Int. Conf. Multimedia*, Oct. 2023, pp. 8580–8589.

[153] J. Shi, H. Zhang, D. Zhou, and Z. Zhang, "Toward intelligent interactive design: generation framework based on cross-domain fashion elements," in *Proc. 31st CM Int. Conf. Multimedia*, Oct. 2023, pp. 7152–7163.

[154] Z. Su, S. Zhao, H. Zhang, P. Li, and Y. Lu, "Digital printing image generation method based on style transfer," *Textile Res. J.*, vol. 93, nos. 23–24, pp. 5211–5223, Dec. 2023.

[155] S. Zhang et al., "Garment ligner: Text-to-garment generation via retrieval-augmented multi-level corrections," in *Proc. Eur. Conf. Comput. Vis.*, 2024, pp. 148–164.

[156] X. Zhang et al., " RM NI: Part-level garment-text alignment for unified cross-modal fashion design," in *Proc. 30th CM Int. Conf. Multimedia*, Oct. 2022, pp. 4525–4535.

[157] Z. Zhang et al., "UFC-BERT: Unifying multi-modal controls for conditional image synthesis," in *Proc. dv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 27196–27208.

[158] S. Zhu, S. Fidler, R. Urtasun, D. Lin, and C. C. Loy, "Be your own prada: Fashion synthesis with structural coherence," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 1689–1697.

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[159] Z. Chong, X. Zhang, F. Zhao, Z. Xie, and X. Liang, "Fashion matrix: Editing photos by just talking," 2023, *arXiv:2307.13240*.

[160] T. Wang and M. Ye, "TexFit: Text-driven fashion image editing with diffusion models," in *Proc. I Conf. rtif. Intell.*, vol. 38, 2024, pp. 10198–10206.

[161] X. Zhang et al., "DiffCloth: Diffusion based garment synthesis and manipulation via structural cross-modal semantic alignment," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 23097–23106.

[162] L. Qiu et al., "RichDreamer: generalizable normal-depth diffusion model for detail richness in text-to-3D," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 9914–9925.

[163] Z. Su, T. Yu, Y. Wang, and Y. Liu, "DeepCloth: Neural garment representation for shape and style editing," *IEEE Trans. Pattern nal. Mach. Intell.*, vol. 45, no. 2, pp. 1581–1593, Feb. 2023.

[164] L. Qiu, G. Chen, J. Zhou, M. Xu, J. Wang, and X. Han, "REC-MV: REconstructing 3D dynamic cloth from monocular videos," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 4637–4646.

[165] M. Sun et al., "IF-garments: Reconstructing your intersection-free multi-layered garments from monocular videos," in *Proc. 32nd CM Int. Conf. Multimedia*, Oct. 2024, pp. 6588–6597.

[166] R. Li, C. Dumery, B. Guillard, and P. Fua, "Garment recovery with shape and deformation priors," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 1586–1595.

[167] Y. Li, F. Luo, and C. Xiao, "Diffusion-FOF: Single-view clothed human reconstruction via diffusion-based Fourier occupancy field," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 9525–9534.

[168] Z. Luo et al., "GarVerseLOD: High-fidelity 3D garment reconstruction from a single in-the-wild image using a dataset with levels of details," 2024, *arXiv:2411.03047*.

[169] . Srivastava, C. Pokhariya, S. S. Jinka, and . Sharma, "XCloth: Extracting template-free textured 3D clothes from a monocular image," in *Proc. 30th CM Int. Conf. Multimedia*, Oct. 2022, pp. 2504–2512.

[170] S. Bang, M. Korosteleva, and S.-H. Lee, "Estimating garment patterns from static scan data," *Comput. Graph. Forum*, vol. 40, no. 6, pp. 273–287, Sep. 2021.

[171] Y. Jeong, K. Hong, and S.-J. Kim, "3D pattern construction and its application to tight-fitting garments for comfortable pressure sensation," *Fibers Polym.*, vol. 7, no. 2, pp. 195–202, Jun. 2006.

[172] W. Ji, X. Li, F. Wu, Z. Pan, and Y. Zhuang, "Human-centric clothing segmentation via deformable semantic locality-preserving network," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 12, pp. 4837–4848, Dec. 2020.

[173] J. Jung, H. Kim, and J. Park, "Deep fashion designer: Generative adversarial networks for fashion item generation based on many-to-one image translation," *Electronics*, vol. 14, no. 2, p. 220, Jan. 2025.

[174] D. Song et al., "Image-based virtual try-on: survey," *Int. J. Comput. Vis.*, vol. 133, no. 5, pp. 2692–2720, May 2025.

[175] M. Nazari, H. Emami, R. Rabiei, H. R. Rabiee, . Salari, and H. Sadr, "Enhancing cardiac function assessment: Developing and validating a domain adaptive framework for automating the segmentation of echocardiogram videos," *Computerized Med. Imag. Graph.*, vol. 124, Sep. 2025, rt. no. 102627.

[176] Y. Jiang, D. Ji, Z. Han, and M. Zwicker, "SDFDiff: Differentiable rendering of signed distance fields for 3D shape optimization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 1248–1258.

[177] M. Feng, L. Zhang, X. Lin, S. Z. Gilani, and . Mian, "Point attention network for semantic segmentation of 3D point clouds," *Pattern Recognit.*, vol. 107, Nov. 2020, rt. no. 107446.

[178] D. Liu et al., "Uni3D-LLM: Unifying point cloud perception, generation and editing with large language models," 2024, *arXiv:2402.03327*.

[179] H. Keshavarz, H. Sadr, M. Nazari, and . Salari, "Facilitating timely decision-making in healthcare: n object detection approach for automated coronary artery stenosis detection," *Frontiers Biomed. Technol.*, vol. 14, pp. 1–12, Jul. 2025.

[180] H. Yan, H. Zhang, J. Shi, J. Ma, and X. Xu, "Toward intelligent fashion design: texture and shape disentangled generative adversarial network," *CM Trans. Multimedia Comput., Commun., ppl.*, vol. 19, no. 3, pp. 1–23, ug. 2023.

[181] K. k, . Kassim, J.-H. Lim, and J. Y. Tham, " ttribute manipulation generative adversarial networks for fashion images," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 10540–10549.

[182] C. Kong, D. Jeon, O. Kwon, and N. Kwak, "Leveraging off-the-shelf diffusion model for multi-attribute fashion image manipulation," in *Proc. IEEE/CVF Winter Conf. ppl. Comput. Vis. (W CV)*, Jan. 2023, pp. 848–857.

[183] K. Sun, P. Zhang, J. Zhang, J. Tao, and K. Yuan, "PFNet: ttribute-aware personalized fashion editing with explainable fashion compatibility analysis," *Inf. Process. Manage.*, vol. 61, no. 1, Jan. 2024, rt. no. 103540.

[184] . S. M. Sayem, "Digital fashion innovations for the real world and metaverse," *Int. J. Fashion Design, Technol. Educ.*, vol. 15, no. 2, pp. 139–141, May 2022.

[185] J.-M. Lu and M.-J. Wang, " computer-aided production system for mass customization in fashion," *Sci. J. Riga Tech. Univ. Comput. Sci.*, vol. 43, no. 1, pp. 104–109, Jan. 2011.

[186] R. Nayak and R. Padhye, *Garment Manufacturing Technology*. msterdam, The Netherlands: Elsevier, 2015, pp. 1–498.

[187] Z. Jiang, J. Guo, and X. Zhang, "Fast custom apparel design and simulation for future demand-driven manufacturing," *Int. J. Clothing Sci. Technol.*, vol. 32, no. 2, pp. 255–270, Nov. 2019.

[188] L. Liu, Z. Su, X. Fu, L. Liu, R. Wang, and X. Luo, " data-driven editing framework for automatic 3D garment modeling," *Multimedia Tools ppl.*, vol. 76, no. 10, pp. 12597–12626, May 2017.

[189] B. Jiang, J. Zhang, H. C. Yang, J. Luo, L. Liu, and H. Bao, "BCNet: Learning body and cloth shape from a single image," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 18–35.

[190] G. Moon, H. Nam, T. Shiratori, and K. M. Lee, "3D clothed human reconstruction in the wild," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 1–25.

[191] . jith et al., "Robotic automation in apparel manufacturing: novel approach to fabric handling and sewing," 2025, *arXiv:2503.00249*.

[192] Y. Jhanji, "Computer-aided design–garment designing and patternmaking," in *utomation in Garment Manufacturing*, R. Nayak and R. Padhye, Eds., Sawston, C , US : Woodhead Publishing, 2018, pp. 253–290.

[193] X. Cong and W. Zhang, "The application of hierarchical perception technology based on deep learning in 3D fashion design," *Heliyon*, vol. 10, no. 9, May 2024, rt. no. e29983.

[194] Z. Wang, "3D representation methods: survey," 2024, *arXiv:2410.06475*.

[195] H. Chen, Y. Yao, and J. Zhang, "Neural- BC: Neural parametric models for articulated body with clothes," *IEEE Trans. Vis. Comput. Graphics*, vol. 31, no. 2, pp. 1478–1495, Feb. 2025.

[196] B. Guillard, F. Stella, and P. Fua, "MeshUDF: Fast and differentiable meshing of unsigned distance field networks," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 576–592.

[197] L. De Luigi, R. Li, B. Guillard, M. Salzmann, and P. Fua, "DrapeNet: Garment generation and self-supervised draping," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 1451–1460.

[198] Y. Li, H.-Y. Chen, E. Larionov, N. Sarafianos, W. Matusik, and T. Stuyck, "Diff vatar: Simulation-ready garment optimization with differentiable simulation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 4368–4378.

[199] S. Rostami, H. Sadr, . shraf, and M. Nazari, " n efficient hybrid deep neural network for intelligence quotient detection from brain magnetic resonance imaging," in *Proc. 11th Int. Conf. Web Res. (ICWR)*, pr. 2025, pp. 515–521.

[200] X. Yu, S. Zhao, S. Luo, G. Yang, and L. Shao, "DiffCloth I: Differentiable cloth simulation with intersection-free frictional contact and differentiable two-way coupling with articulated rigid bodies," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Oct. 2023, pp. 400–407.

[201] Y. Li, T. Du, K. Wu, J. Xu, and W. Matusik, "DiffCloth: Differentiable cloth simulation with dry frictional contact," *CM Trans. Graph.*, vol. 42, no. 1, pp. 1–20, Oct. 2022.

[202] D. Baraff and . Witkin, "Large steps in cloth simulation," in *Proc. 25th nnu. Conf. Comput. Graph. Interact. Techn. (SIGGR PH)*, 1998, pp. 43–54.

[203] Y. Yunchu and Z. Weiyuan, "Prototype garment pattern flattening based on individual 3D virtual dummy," *Int. J. Clothing Sci. Technol.*, vol. 19, no. 5, pp. 334–348, Oct. 2007.

[204] M. Zhuang and X. Zhang, " novel apparel surface flattening algorithm," *Int. J. Clothing Sci. Technol.*, vol. 25, no. 4, pp. 300–316, Jul. 2013.

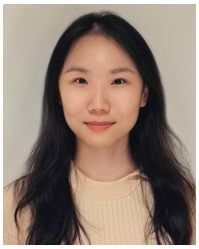
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